

## Research

# Utilizing fuzzy logic, particle swarm optimization, desirability function and NSGA-II for multi-objective optimization of sponge Gourd–Bagasse polymer composite properties

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## Abstract

The process of developing mathematical models to forecast material properties can be intricate and time-consuming. Consequently, there has been a shift towards simpler and more precise computational models founded on artificial intelligence. In this particular research, fuzzy logic (FL) was employed to anticipate the mechanical properties of a newly formulated hybrid composite material, which combines sponge gourd, bagasse, and epoxy resin for golf club applications. Furthermore, a multi-objective optimization of these properties was conducted using a modified desirability function (DF) and the Non-Dominated Sorting Genetic Algorithm (NSGA-II). The FL model took into account three input factors: the weight percentage of bagasse, the weight percentage of sponge gourd, and the fiber size measured in micrometers. The response variables examined were tensile strength, hardness, flexural strength, modulus, elongation, and impact strength. The FL model was combined with the modified DF algorithm and the NSGA-II algorithm, respectively. To optimize the DF, the particle swarm optimization (PSO) algorithm was employed. The results demonstrated that the FL model accurately predicted the mechanical properties of the hybrid composite material. The lowest correlation coefficient (R) between experimental responses and FL predictions was found to be 0.9529. The modified algorithms addressed specific characteristics in the desirability properties, such as elongation, where the desirability remained constant within a range. When the modified algorithm was utilized, the optimized properties exhibited a desirability value of 0.84, surpassing the values of 0.83 and 0.82 obtained without modifying the optimization algorithms. In conclusion, the FL model in conjunction with the modified DF algorithm and the NSGA-II algorithm successfully predicted the mechanical properties of the hybrid composite material. The modifications made to the algorithms contributed to enhancing the optimization process of these properties.

**Keywords** Fuzzy logic · Genetic algorithm · Particle swarm optimization · Desirability function · Composite · Multi objective optimization

## Abbreviations

|    |                  |
|----|------------------|
| Bg | Bagasse          |
| Sg | Sponge gourd     |
| Fs | Fibre size       |
| Ts | Tensile strength |

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|         |                                         |
|---------|-----------------------------------------|
| H       | Hardness                                |
| FSt     | Flexural strength                       |
| AHP     | Analytical hierarchical process         |
| PSO     | Particle swarm optimization             |
| FL      | Fuzzy logic                             |
| MOD     | Modulus                                 |
| EI      | Elongation                              |
| Is      | Impact strength                         |
| Sstr    | Specific strength                       |
| Sstif   | Specific stiffness                      |
| Des     | Desirability                            |
| NSGA-II | Non-dominated sorting genetic algorithm |
| DF      | Desirability function                   |
| ANN     | Artificial neural networks              |

## 1 Introduction

In recent years, the development of sustainable materials has gained significant attention due to the pressing need for eco-friendly solutions in various industries [1]. Polymer composites, in particular, have emerged as a promising alternative to conventional materials, offering enhanced mechanical, thermal, and environmental properties [2]. Among the diverse range of natural fibers available for reinforcing polymer composites, sponge gourd-bagasse has shown great potential due to its abundance, renewability, and biodegradability. However, optimizing the properties of sponge gourd-bagasse polymer composites presents a complex challenge. These materials possess multiple performance characteristics, such as tensile strength, hardness, flexural strength, modulus, elongation, and impact strength, which often conflict with one another. Achieving the desired balance between these properties requires a comprehensive approach that considers the synergistic effects of various parameters and optimizes multiple objectives simultaneously.

To address this optimization problem, this paper proposes a novel framework, which is largely based on artificial intelligence and soft computing techniques, and combines fuzzy logic, particle swarm optimization (PSO), the desirability function, and the Non-Dominated Sorting Genetic Algorithm II (NSGA-II). Fuzzy logic allows for the representation of imprecise and uncertain information, enabling a more realistic modeling of the complex relationships between the composite's constituent materials, processing parameters, and resulting properties. PSO, a metaheuristic algorithm inspired by the social behavior of birds or insects, provides an efficient approach to searching the parameter space for optimal solutions [3]. The desirability function facilitates the aggregation of multiple objectives into a single measure of desirability, enabling the simultaneous optimization of diverse properties. NSGA-II, a widely used multi-objective evolutionary algorithm, allows for the exploration of the Pareto-optimal front, enabling decision-makers to choose the most suitable compromise solution. By harnessing the power of these AI based models we are empowered to anticipate a wide range of outcomes, including properties of materials, outputs of chemical processes, and outputs of industrial processes. Prominent research studies conducted by Umezuegbu et al. [4], Onukwuli et al. [5], Sinha et al. [6], Nwobi-Okoye and Umeonyiagu [7], and Atuanya et al. [8] emphasize the significance of models and their diverse applications across different domains.

It is worth noting that combining NSGA-II and desirability function can lead to significant improvements in material design and development since NSGA-II efficiently generates a set of Pareto-optimal solutions, while the desirability function provides a mechanism to rank and select the most preferred solution based on specific preferences. The synergy between NSGA-II and the desirability function can lead to the identification of high quality solutions that meet multiple criteria.

## 2 Related works

The progress of modern engineering heavily relies on the creation of precise prediction and optimization models tailored for materials science and engineering applications. As a result, extensive research efforts are currently concentrated in this area, with artificial intelligence (AI) playing a prominent role [9–12]. AI encompasses a diverse range of tools utilized in the realms of science and engineering, such as Artificial Neural Networks (ANN), Adaptive Neuro-Fuzzy Inference System (ANFIS), FL, Genetic Algorithm (GA), and PSO [13–17].

In the specific application being considered, Fuzzy Logic, NSGA-II and PSO were employed to predict and optimize the properties of a composite material intended for golf club applications. Extensive scientific literature provides a wide range of examples highlighting the diverse applications of artificial intelligence (AI) in predicting and optimizing materials properties. Several studies [8, 18–23] have explored this area. For example, in their study, Nwobi-Okoye et al. [13] conducted a comparison between the predictive performance of ANN and adaptive neuro-fuzzy inference system (ANFIS) for hardness and cost outputs in the age hardening process. ANFIS demonstrated superior performance in hardness prediction, while both models achieved similar levels of accuracy in cost prediction. ANN, with better performance outside the experimental data points, was then used in multi-objective optimization of the process parameters using a genetic algorithm (GA). The study concluded that ANN, with less stringent data requirements, is more effective and cost-efficient in predicting process outputs and optimizing the age hardening operation.

In a similar vein, the research conducted by Nwobi-Okoye et al. [14] centered on the modeling and optimization of parameters for dry foundry sand using ANN and genetic algorithm (GA). The ANN exhibited impressive predictive capabilities, achieving a correlation coefficient of 0.99116 for dry compressive strength and a correlation coefficient of 1 for unit production cost. The trained ANN was then used as the fitness function for the GA, enabling multi-objective optimization of compressive strength and production cost, leading to the identification of the Pareto front and determining the optimal process input parameters for achieving the desired strength and cost.

Furthermore, Chowdhury et al. [24] used fuzzy logic for predicting and optimizing the erosion rate of carbon fiber-reinforced ebonite. This approach demonstrated the effective utilization of fuzzy logic in this context. Through the analysis of experimental data, a fuzzy logic model was developed to accurately predict the erosion rate by taking into account variations in impact velocity, impingement angle, erodent size, and stand-off distance. The model exhibited a remarkable precision of 94.45%. In an independent research investigation, Sinha et al. [6] employed fuzzy logic to forecast the mechanical properties of a hybrid composite material reinforced with abaca fiber. The prediction model took into account variables such as the weight percentage of abaca fiber, particle size of red mud, and weight percentage of red mud, while the output parameters focused on tensile strength and flexural strength. The model achieved a commendable prediction accuracy of 87%. Furthermore, Umeonyiagu and Nwobi-Okoye [15] employed Artificial Neural Networks and NSGA-II to carry out multi-objective optimization of bamboo-reinforced concrete beams. Their optimization objectives were twofold: minimizing cost while simultaneously maximizing the tensile and flexural strengths of the concrete beams. The input parameters taken into consideration were the percentage of bamboo composition and age hardening days. The findings of this study highlighted that the optimization results served as valuable guidance in designing resilient and cost-effective concrete beams reinforced with bamboo. In a distinct investigation, Nwobi-Okoye and Uzochukwu [17] conducted a multi-objective optimization study on Al 631/Egg shell metal matrix composite material produced through the stir casting process. This study focuses on the modeling and multi-objective optimization of Al 6351/Egg Shell Reinforced Composite, specifically targeting toughness and hardness properties. Response Surface Methodology (RSM) and ANN were utilized to model and predict the composite properties based on variations in production process variables. The ANN model was further employed as the fitness function in the NSGA-II for multi-objective optimization. The results demonstrate the effectiveness of RSM in modeling the composite properties, while the ANN predictions exhibited excellent correlation coefficients with experimental responses. The optimized Pareto front obtained through the NSGA-II algorithm provides valuable insights for practical applications of the Al 6351/Egg Shell Reinforced Composite in engineering design. Nwobi-Okoye et al. [16] conducted a study in which they employed the NSGA-II algorithm in conjunction with an ANN-based prediction model as the objective function for multi-objective optimization of particle boards. The primary objective of this research is to model and perform multi-objective optimization of a composite material made from Newbouldia Laevies Fibre and recycled high-density polyethylene (RHDPE), specifically for fiberboard applications. To achieve this, Response Surface Methodology (RSM), ANN, and NSGA-II were utilized as the chosen methodologies. The properties of the fiberboard, including modulus of rupture (MOR), modulus of elasticity (MOE), internal bonding (IB), thickness

swelling (TS), and water absorption (WA), were optimized by considering various production process variables. The RSM and ANN models effectively captured the variations in physical and mechanical properties, while the NSGA-II algorithm provided an optimized Pareto front that serves as a practical design guide for composite applications. The study demonstrated the superiority of the NSGA-II algorithm over the traditional desirability function for multi-objective optimization in this context. In a different research endeavor, Tarasov et al. [25] developed a fuzzy logic-based prediction model to accurately represent the yield strength of as-cast A356 alloy. This paper tackles the issue of handling uncertain and imprecise data in domains like casting lightweight components. The authors present a novel approach based on fuzzy logic to model the fatigue-related mechanical properties of as-cast components, which is an area that has received limited attention in previous research. Specifically, they develop two fuzzy logic models that relate yield strength to the chemical composition and rate of solidification of castings for two A356 alloys, and compare their performance to artificial neural networks. The comparison shows that although neural networks achieve similar prediction accuracy, the fuzzy logic approach is deemed more appropriate for the domain due to its interpretability and explainability. The fuzzy logic models exhibit prediction errors at a similar level as reported in related work, and these errors are influenced by the parameters of the membership functions in the model. Overall, this research provides a valuable contribution to the understanding and modeling of fatigue-related mechanical properties in cast components, particularly in the context of lightweight castings. In their research, Aamir et al. [26] combined the Taguchi Method and Fuzzy Logic to model and optimize process parameters in multi-hole simultaneous drilling. The paper focuses on optimizing drilling parameters for industries like aerospace and automotive, where precise geometric tolerances are critical. The Taguchi method is employed to optimize drilling parameters for both one-shot drilling and multi-hole drilling processes. Analysis of variance and regression analysis are utilized to evaluate the significance of drilling parameters and their impact on surface roughness and hole size. Furthermore, a fuzzy logic approach is employed to predict surface roughness and hole size, resulting in excellent agreement with experimental values. Confirmation tests validate the efficacy of the Taguchi optimization and fuzzy models in accurately representing surface roughness and hole size in multi-hole drilling applications.

The study of Nwobi-Okoye and Ochieze [27] focused on the development of a cost-effective and lightweight ceramic-based aluminum metal matrix composite (MMC) by incorporating cow horn particles (CHp) into an aluminum alloy A356. The objective was to model the age hardening process of the composite using response surface methodology (RSM) and artificial neural network (ANN), and then optimize the process parameters using a simulated annealing optimization algorithm (SA-NN system). The results indicate that the ANN model outperformed the RSM in accurately predicting the age hardening data, with a higher correlation coefficient of 0.9921 compared to 0.9583 for RSM. The SA-NN system effectively optimized the process parameters, producing values that closely matched the experimental results. The maximum relative error was 1.2%, the minimum was 0.35%, and the average was 0.71%.

Fitas et al. [28] optimized composite structures to save costs and improve structural safety by employing the Finite Element Method for evaluation. To achieve this, the study combined Particle Swarm Optimization with techniques from Genetic Algorithms. The enhanced optimization method was applied to composite structures using Finite Element Analysis to evaluate performance. Results demonstrated that the proposed approach effectively generated a larger set of optimal design solutions, leading to lighter and more robust composite structures compared to traditional methods.

Based on the fact that AI and data science have accelerated materials development, but predicting optimal material properties is insufficient, Zhang et al. [29] focused their research on determining the next experimental steps to refine material design. The authors proposed an improved genetic algorithm, NSGA-IIP, which incorporates property preferences into the optimization process. By considering both convergence and diversity of solutions while aligning with desired properties, NSGA-IIP enhances material design efficiency. The algorithm's effectiveness is demonstrated through application to thermal barrier ceramic coatings, outperforming previous methods in terms of practicality and time efficiency. Sinha et al. [30], investigated hybrid abaca-epoxy composites reinforced with red mud as a potential eco-friendly material for tribological applications. The study utilized response surface methodology (RSM) to determine the influence of abaca and red mud content, as well as red mud particle size, on the sliding wear of the composites. An optimal composite composition for minimum sliding wear was identified through RSM. Additionally, a fuzzy logic model was developed to predict sliding wear, demonstrating an accuracy of 87% when validated with experimental data. This research contributes to the understanding of hybrid composite wear behavior and provides a tool for predicting wear performance.

This present study distinguishes itself from the reviewed literature by adopting a unique methodology. The study implemented a customized desirability function, which was optimized using Particle Swarm Optimization (PSO) while employing a fuzzy logic model as its fitness function. Furthermore, an adapted NSGA-II algorithm, coupled with a fuzzy

**Fig. 1** Untreated fiber from baggase



**Fig. 2** Untreated fiber from sponge gourds



logic prediction model, was employed as an additional fitness function. The objectives of this study are twofold: (1) to develop a comprehensive model that incorporates fuzzy logic to represent the complex relationships between the composite's constituent materials, processing parameters, and properties, and (2) to employ a hybrid optimization approach that combines PSO, the desirability function, and NSGA-II for multi-objective optimization of sponge gourd-baggasse polymer composite properties.

By utilizing this integrated approach, this research aims to enhance the understanding of the interdependencies between the various parameters and optimize the mechanical, thermal, and environmental properties of sponge gourd-baggasse polymer composites. The outcomes of this study are expected to contribute significantly to the development of sustainable materials with improved performance characteristics, thereby fostering their utilization in various industrial applications and promoting a more sustainable future.

### 3 Methodology

#### 3.1 Composite material production

##### 3.1.1 Materials

For this study, the bagasse fiber employed, as shown in Fig. 1, was obtained from the local market in Awka, Nigeria. Likewise, the sponge gourd depicted in Fig. 2 was locally sourced from the nearby region. The required epoxy resin and hardener were procured from the local market specifically for use in the experimental procedures.

##### 3.1.2 Preparation of the fibers

The bagasse and sponge gourd fibers were prepared separately and through independent processes. First, the bagasse fiber underwent several steps. It was soaked in water for 24 h to eliminate any dirt or debris present in the fiber, followed by sun drying for 2 days. Subsequently, the dried bagasse fiber was soaked in a 10% concentration NaOH solution for

**Fig. 3** Treatment of the fiber using dilute NaOH solution



24 h (as depicted in Fig. 3). This treatment aimed to remove impurities such as lignin and wax, thereby enhancing the fiber's roughness and adhesive properties. The alkaline-treated fiber was then rinsed with water until a neutral pH value of 7 was achieved, ensuring a neutral fiber solution. After rinsing, the fiber was sun dried for an additional 2 days before undergoing methanol treatment. During methanol treatment, the fiber was soaked in a dilute methanol solution (60:40) for 3 h, further enhancing its roughness and bonding characteristics, as shown in Fig. 4. Following methanol treatment, the fiber was sun dried for 2 days, preparing it for subsequent pulverization. Similar procedures were repeated for the preparation of sponge gourd fiber.

### 3.1.3 Particle size analysis

To increase the surface areas of both bagasse and sponge gourd fibers, a pulverization technique was employed. This technique effectively converted the long grain fibers into a powdered form, as depicted in Fig. 5, enhancing their surface area for subsequent use.

To determine the particle size of the pulverized fibers, the analysis followed the guidelines specified by ASTM-60 standards. Approximately 200 g of the fiber particles were carefully placed into a series of sieves arranged in a descending order of fineness. The sieves were then subjected to electrical shaking for a recommended duration of 15 min, ensuring a comprehensive classification process. This procedure, as depicted in Fig. 6, facilitated the measurement of the weights of particles retained on the 150, 300, and 600  $\mu\text{m}$  sieves, which were utilized for the research purposes.

### 3.1.4 Composite sample production

Using an airtight electronic weighing balance, as depicted in Fig. 7, the weight percentages of bagasse and sponge gourd fibers, along with the polymer, were precisely measured. These measurements were derived from the data analysis obtained through the Taguchi experimental design.

**Fig. 4** Fiber treatment with methanol



**Fig. 5** Fibers in pulverized form. **a** Bagasse fiber in pulverized form. **b** Sponge gourd fiber in pulverized form



(a). Bagasse fiber in pulverized form

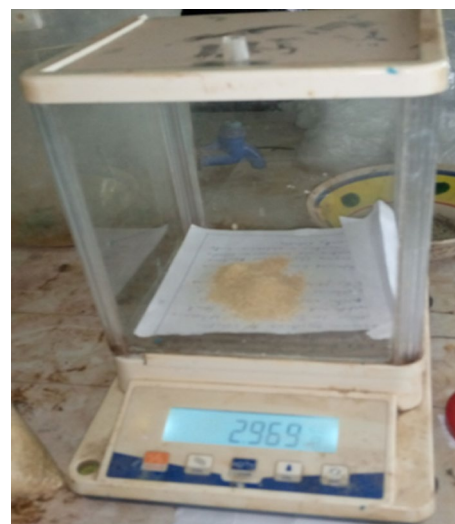


(b). Sponge gourd fiber in pulverized form

**Fig. 6** The electronic sieving machine



**Fig. 7** The electronic weighing machine



To ensure smooth casting, the paraffin wax was heated to its melting point (50 °C) using a Bunsen burner. The liquefied wax was then promptly applied to the metal mold, enabling easy removal of the samples once cast. Next, the composite materials, consisting of the predetermined weight percentages of bagasse, sponge gourd, polymer, and hardener, were meticulously mixed in a crucible plate. Continuous stirring with a stirrer was employed to achieve a homogeneous blend.

**Table 1** Factors and there levels

| Factor                  | Level |
|-------------------------|-------|
| Wt% of baggase (A)      | 3     |
| Wt% of sponge gourd (B) | 3     |
| Fiber size (C)          | 3     |

**Table 2** Degree of freedom (D.O.F)

| Factors | Level (S) | D.O.F = S-1 |
|---------|-----------|-------------|
| A       | 3         | 2           |
| B       | 3         | 2           |
| C       | 3         | 2           |
| A×B     |           | 2×2=4       |
| A×C     |           | 2×2=4       |
| B×C     |           | 2×2=4       |
|         |           | D.O.F = 18  |

The resulting mixture was carefully poured into a waxed metal mold with dimensions of 300 mm × 300 mm and allowed to solidify. The production of the composite material followed the specific design configuration outlined in Table 5.

3.1.5 Design of experiment

Among experimental design techniques for material production optimization, full factorial design [31] and Taguchi robust design method are quite popular [32]. The experimental design utilized in this study followed the Taguchi robust design method, which aims to reduce process variation and achieve high-quality products while minimizing costs for manufacturers. Genichi Taguchi developed this method to investigate how different parameters affect the average and variation of a process performance characteristic, which reflects the process's effectiveness.

The Taguchi experimental design approach involves organizing the process parameters and their corresponding levels using orthogonal arrays. Unlike factorial design, which requires testing all possible combinations, the Taguchi method focuses on testing pairs of combinations. This strategy enables the collection of essential data to identify the most influential factors affecting product quality, while minimizing the number of experiments required. As a result, time and resources are saved.

Conventional experimental design methods can be complex and challenging, especially when dealing with a large number of process parameters. To address this issue, the Taguchi method utilizes specially designed orthogonal arrays to explore the entire parameter space with a reduced number of experiments.

In this study, three key factors related to superplastic forming were considered: the weight percentage of fiber 1 (baggase), the weight percentage of fiber 2 (sponge gourd), and the fiber particle size. Each factor was examined at three levels: low, medium, and high, denoted by the values 1, 2, and 3, respectively [33].

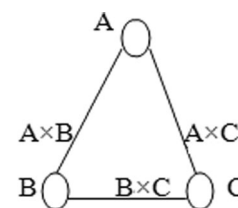
Orthogonal arrays are unique standard experimental designs that enable the identification of the main factors influencing the output using a minimal number of experimental trials [33]. When selecting an orthogonal array, it is essential to follow the following steps:

Step 1: Identify the factors that have three distinct levels. The factors and their corresponding levels are presented in Table 1

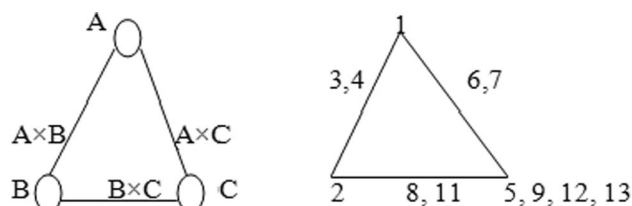
Step 2: Calculate the degrees of freedom (DOF)

The degrees of freedom (DOF) quantify the number of independent variations that a system can undergo under a given constraint. The specific degrees of freedom for each factor are indicated in Table 2.

**Fig. 8** Required linear graph for  $L_{27}$



**Fig. 9** Required linear graph and standard linear graph for  $L_{27}$



### Step 3: Determining the total number of experimental runs

In this particular step, the calculation of the total number of experimental runs is determined by considering the factors and levels incorporated in the chosen orthogonal array. The precise calculation methodology varies depending on the configuration and type of orthogonal array selected.

$$T_{\text{experiment}} = 1 + \text{D.O.F} = 1 + 18 = 19$$

### Step 4: Selecting a suitable orthogonal array

In this step, an orthogonal array is chosen that meets the requirement of having a size greater than or equal to 19. After consulting the standard orthogonal array table, it was determined that the nearest available orthogonal array is  $L_{27}$ .

### Step 5: Visualizing the required linear graph (RLG)

To aid in the analysis and interpretation of the experimental data, a Required Linear Graph (RLG) was constructed. The RLG is depicted in Fig. 8, providing a visual representation of the experimental design and its corresponding factors and levels.

### Step 6: Comparing the required linear graph and standard linear graph for $L_{27}$

In this step, the required linear graph (RLG) obtained from the experiment is compared to the standard linear graph corresponding to the chosen  $L_{27}$  orthogonal array. The purpose of this comparison is to assess the alignment between the actual experimental results and the expected outcomes based on the standard array. The comparison is visually represented in Fig. 9.

The Required Linear Graph (RLG) provides a representation of the factors and their interactions. When comparing the RLG to the Standard Linear Graph (SLG), it is observed that factors A, B, and C correspond to columns 1, 2, and 5 in the SLG. Meanwhile, columns 3 and 4, 8 and 11, as well as 6 and 7, are omitted as they represent the confounded interactions between columns 1 and 2, columns 2 and 5, and columns 1 and 5, respectively.

### Step 7: Allocating the factors to the columns of the orthogonal array is illustrated in Table 3.

Step 8: The experimental layout is determined by selecting the appropriate experimental design layer from the default orthogonal array design, as illustrated in Table 4.

**Table 3** Factors with node assigned

| Factors | Column/<br>node<br>assigned |
|---------|-----------------------------|
| A       | 1                           |
| B       | 2                           |
| C       | 5                           |
| A×B     | 3, 4                        |
| B×C     | 8, 11                       |
| A×C     | 6, 7                        |

**Table 4** Default orthogonal array

| Factor                  | Level 1 | Level 2 | Level 3 |
|-------------------------|---------|---------|---------|
| Wt% of baggase (A)      | 5 wt%   | 10 wt%  | 20 wt%  |
| Wt% of sponge gourd (B) | 5 wt%   | 10 wt%  | 20 wt%  |
| Fiber size (C)          | 150 µm  | 300 µm  | 600 µm  |

**Table 5** Number of experimental runs with their parameters and data

| Factors |             |                  |                 |
|---------|-------------|------------------|-----------------|
| S/N     | %wt bagasse | %wt Sponge Gourd | Fiber size (µm) |
| 1       | 5           | 5                | 600             |
| 2       | 10          | 5                | 600             |
| 3       | 20          | 5                | 600             |
| 4       | 5           | 10               | 600             |
| 5       | 10          | 10               | 600             |
| 6       | 20          | 10               | 600             |
| 7       | 5           | 20               | 600             |
| 8       | 10          | 20               | 600             |
| 9       | 20          | 20               | 600             |
| 10      | 5           | 5                | 300             |
| 11      | 10          | 5                | 300             |
| 12      | 20          | 5                | 300             |
| 13      | 5           | 10               | 300             |
| 14      | 10          | 10               | 300             |
| 15      | 20          | 10               | 300             |
| 16      | 5           | 20               | 300             |
| 17      | 10          | 20               | 300             |
| 18      | 20          | 20               | 300             |
| 19      | 5           | 5                | 150             |
| 20      | 10          | 5                | 150             |
| 21      | 20          | 5                | 150             |
| 22      | 5           | 10               | 150             |
| 23      | 10          | 10               | 150             |
| 24      | 20          | 10               | 150             |
| 25      | 5           | 20               | 150             |
| 26      | 10          | 20               | 150             |
| 27      | 20          | 20               | 150             |

Table 5 presents the number of experimental runs along with their corresponding parameters. The responses measured in this study included tensile strength, hardness, flexural strength, flexural modulus, elongation, and impact strength. The Table indicates that a total of 27 experiments were conducted following the prescribed Taguchi experimental design.

### 3.1.6 Composite material testing procedure

Several properties were tested in this study, including density, tensile strength, hardness, modulus, flexural strength, elongation, and impact strength.

#### 3.1.7 Density

To determine the density of the composite material, mathematical expressions and calculations were employed. The composite was poured into a small graduated cylinder with a known volume, and the density was subsequently derived from the obtained measurements.

Let volume of the cylinder =  $V$ .

Mass of cylinder =  $M_1$ .

Mass of cylinder + composite =  $M_2$ .

Therefore,

Mass of composite  $M_3 = M_2 - M_1$ .

Thus density of composite = mass/volume

$$\text{Density} = \frac{M_3}{V} \quad (1)$$

The density was used to calculate the mass of the composite needed to fill a mould of known volume (300 mm × 100 mm × 20 mm). Knowing the volume, the density formula helped determine the exact mass of the composite required. For a mould volume  $V = 300 \times 100 \times 20$ . The density formula allows us to determine the mass of the composite to be poured.

**3.1.7.1 Flexural test** Before conducting the tests, all the samples were subjected to a conditioning process at a temperature of 25°C and a relative humidity of 65%, following the guidelines outlined in ASTM D618-08.

The test samples were prepared in accordance with the prescribed standard for flexural testing. The flexural tests were carried out using a M500-25CT testometric testing machine as depicted in Fig. 10.

For the flexural test, specimens with dimensions of 200 mm × 20 mm × 5 mm were utilized in a three-point bending configuration. The specimens were subjected to a continuous load at a uniform rate of 50 mm/min.

To determine the flexural strength, the test adhered to the standard procedure specified in ASTM D790-10. The flexural strength of each specimen was computed using the following equations:

$$R_b = \frac{3P_{\max}L}{2bd^2} \quad (2)$$

where  $b$  = width of specimen measured in dry condition (mm);  $d$  = thickness of span in mm;  $P_{\max}$  = maximum load (N).

The flexural test was used to measure the bending strength of the composite. The formula was employed to calculate the flexural strength after applying a load to the composite specimen in a three-point bending test setup. By using a specimen of known dimensions (width  $b$  and thickness  $d$ ) and applying a maximum load ( $P_{\max}$ ), the flexural strength ( $R_b$ ) could be calculated to evaluate the composite's resistance to bending.

**3.1.7.2 Tensile strength** The University of Lagos, Nigeria, conducted the tensile strength measurement experiment using a testometric testing machine. To perform the tensile test, composite samples with dimensions of 160 mm × 20 mm × 4 mm were used, following the ASTM 638-10 standard test method for tensile properties of poly-

**Fig. 10** M500-25CT testometric testing machine



mers. The tests were conducted at a constant strain rate of 0.5 mm/min. The maximum tensile strength was determined using the appropriate equation for calculating this value as stated in Eq. (3).

$$R_T = \frac{P_{max}}{BD} \quad (3)$$

where B = width of reduced cross section of the specimen measured in dry conditioning (mm); D = thickness of the specimen measured in dry condition; P<sub>max</sub> = maximum tensile load (N); R<sub>T</sub> = maximum tensile stress (MPa).

The tensile strength testing procedure involved applying a tensile load to the composite specimen and measuring the maximum load it could withstand. The formula was used to compute the tensile strength. With the specimen's width (B) and thickness (d) known, the maximum load (P<sub>max</sub>) from the tensile test was used in the formula to calculate the tensile strength (R<sub>T</sub>), determining the material's ability to handle pulling forces.

**3.1.7.3 Impact strength test** The impact testing of the composite samples was conducted using an Avery dension test machine. The samples were prepared with dimensions of 100 mm × 20 mm × 5 mm, and the Charpy impact tests were performed according to the ASTM D256-10 standard test method for determining the Charpy pendulum impact resistance of plastics. Before conducting the test, the samples were securely mounted on the machine, and the pendulum was released to calibrate the equipment. The test samples were then clamped horizontally in a vice, and the pendulum's swinging motion released the force required to fracture the bar. The angular displacement of the pendulum before the sample broke represented the energy absorbed during the fracture. This value was obtained by reading the calibrated scale on the machine.

The impact strength was determined by subjecting the composite to a sudden force using the Charpy impact test. The energy absorbed during the fracture provided the impact strength. The energy required to break the composite sample was measured, which indicated its toughness and ability to resist sudden impacts.

**3.1.7.4 Hardness test** Hardness testing of the samples was performed at the Department of Metallurgical and Materials Engineering, University of Nigeria, Nsukka, utilizing a Leeb hardness tester. The Leeb hardness tests were conducted on samples with dimensions of 20 mm × 20 mm × 20 mm, following the ASTM A956 standard.

Before conducting the test, the samples were securely mounted on the anvil, as depicted in Fig. 11. The testing procedure involved employing an impact device that utilized spring force to accelerate an impact body. The velocity of the impact body can be divided into three distinct phases.

**Fig. 11** Leeb hardness tester

1. Approach phase: During this initial phase, the spring force accelerates the impact body towards the test surface.
2. Impact phase: In this phase, the impact body makes direct contact with the specimen, resulting in elastic and plastic deformation. The impact body comes to a complete stop while the elastic spring-back of both the impact body and the specimen causes the impact body to rebound.
3. Rebound phase: This final phase involves the acceleration of the impact body out of the impact phase using the remaining energy.

The Leeb hardness (HL) is determined by calculating the ratio of the rebound velocity ( $V_r$ ) to the impact velocity ( $V_i$ ) and multiplying it by a factor of 1000. This ratio provides the Leeb hardness value.

$$HL = \frac{V_r}{V_i} \times 1000 \quad (4)$$

After obtaining the Leeb hardness results in HL, they are converted to the traditional hardness scale HR (hardness according to Rockwell) using a classified hardness chart.

Hardness testing involved measuring the resistance of the composite to indentation using a Leeb hardness tester. The formula was used to calculate the hardness based on the rebound and impact velocities. By recording the impact and rebound velocities, the hardness (HL) of the composite was calculated, helping to determine its wear resistance.

**3.1.7.5 Specific strength** Specific strength, also referred to as the strength-to-weight ratio, is determined by measuring the ratio of strength to the weight of a material or component as stated in Eq. (5).

$$\text{Specific strength (Nm/kg)} = \frac{\text{tensile strength}}{\text{density of the material}} \quad (5)$$

Specific strength was calculated to understand how strong the composite is relative to its weight. During the testing, the tensile strength of the composite was measured, and then the density was used to compute the specific strength. If the tensile strength is known from tensile testing and density is measured, the specific strength formula helps determine the material's efficiency in high-strength, low-weight applications.

**3.1.7.6 Specific stiffness** Specific stiffness, alternatively known as the modulus-to-weight ratio, is quantified as follows:

$$\text{Specific Modulus (Nm/kg)} = \frac{\text{modulus}}{\text{density of the material}} \quad (6)$$

Specific stiffness was assessed to understand the material's stiffness relative to its weight. The modulus (stiffness) of the composite was measured, and the density was used to compute the specific stiffness. By using the measured modulus and density, the specific stiffness was calculated to evaluate the composite's suitability for applications requiring lightweight and stiff materials.

## 3.2 Fuzzy logic modelling

### 3.2.1 Introduction

Fuzzy logic modeling has gained prominence as a valuable approach for predicting the properties of composite materials, offering a robust and flexible framework to tackle the challenges associated with property prediction in complex material systems. In this subsection, we focus on the application of fuzzy logic modeling specifically for the prediction of composite material properties. By harnessing the ability of fuzzy logic to handle imprecise and uncertain data, we can effectively capture the inherent complexities and variations within composite materials to make accurate property predictions.

The objective of this subsection is to provide a comprehensive overview of the fuzzy logic modeling approach for predicting composite material properties. We will delve into the fundamental principles of fuzzy logic, including membership functions, fuzzy rules, and fuzzy inference systems, as they relate to property prediction. Additionally, we will explore the process of constructing a fuzzy logic model specifically tailored to predicting the desired properties of composite materials, considering key input variables, experimental data, and domain knowledge.

By incorporating fuzzy logic modeling into the prediction of composite material properties, we aim to enhance the accuracy and reliability of property estimations. This approach enables us to account for the uncertainties and variations inherent in composite materials, allowing for more informed decision-making and design optimization.

This subsection serves as a guide for researchers and practitioners interested in employing fuzzy logic modeling for predicting composite material properties. By understanding the principles and methodologies presented here, readers will be equipped with the necessary tools to develop their own fuzzy logic models and make reliable predictions of composite material properties.

### 3.2.2 Fuzzy modelling procedure

The data collected from testing the composite material was utilized for fuzzy logic modeling. The input–output data from the experiment, as described in Sect. 3.1.6, underwent both fuzzification and defuzzification processes, as depicted in Fig. 12. These procedures were performed to develop a fuzzy logic model that can predict the properties of the composite material.

The utilization of a fuzzy inference system to predict the mechanical properties of the hybrid composite material under investigation is depicted in Fig. 13. This fuzzy inference system takes as inputs the weight percentages of bagasse, sponge gourd, and fiber size, as illustrated in Fig. 13. Following the process of defuzzification, the system generates outputs that encompass tensile strength, flexural strength, elongation, impact strength, hardness, and modulus.

The membership functions for the weight percentages of bagasse and sponge gourd, as well as the fiber size, are presented in Figs. 14, 15 and 16, respectively. These membership functions are characterized by five linguistic variables: Very Low (VL), Low (LW), Medium (M), High (H), and Very High (VH).

Furthermore, the membership function for fiber size, as shown in Fig. 16, encompasses eight variables denoted as FS1, FS2, FS3, FS4, FS5, FS6, FS7, and FS8.

A total of six fuzzy logic models were employed for prediction modeling. Model I used 24 linguistic variables to predict hardness (Fig. 17a), while Model 2 utilized 49 linguistic variables for flexural strength (Fig. 17b). Model 3 incorporated 26

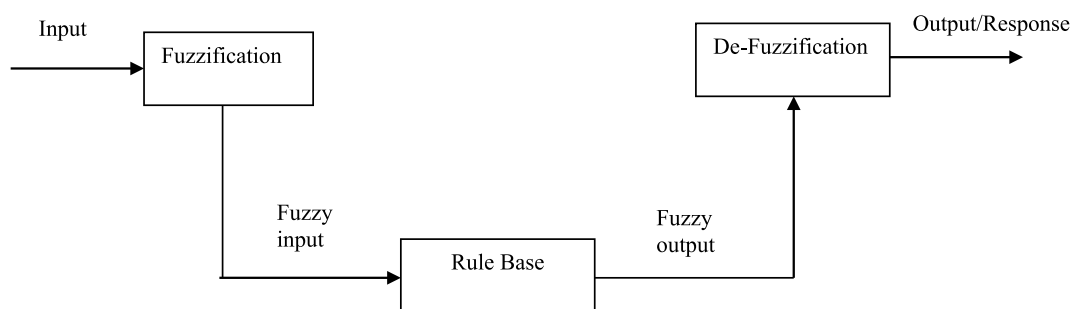


Fig. 12 Component of fuzzy system

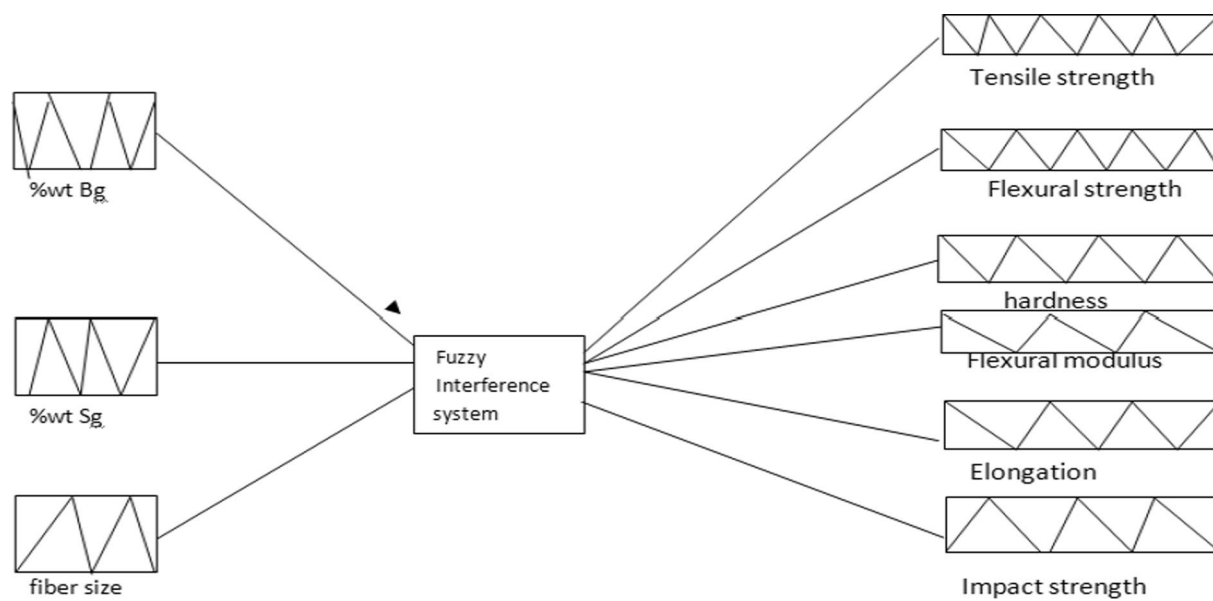


Fig. 13 Fuzzy inference system of hybrid composite

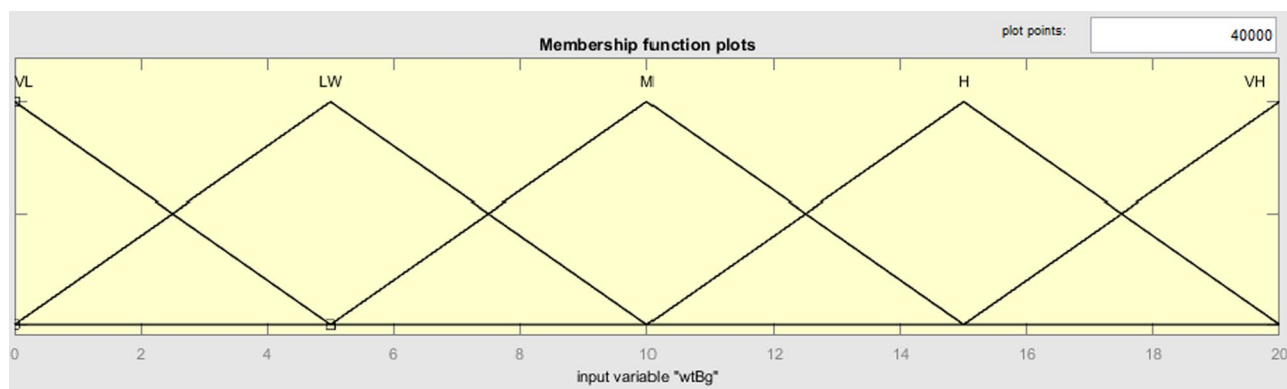


Fig. 14 Membership function for %wt Bagasse

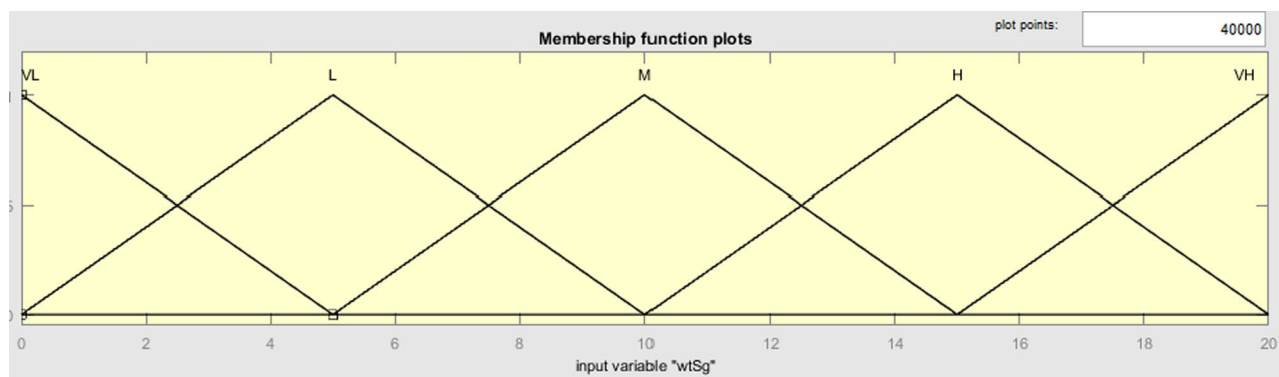


Fig. 15 Membership function for %wt sponge gourd

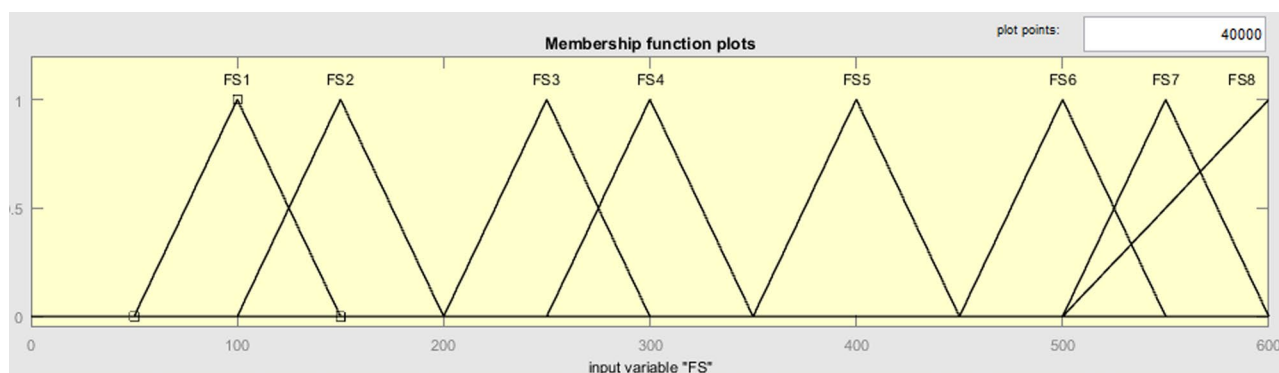


Fig. 16 Membership function for the fiber size

linguistic variables to estimate flexural modulus (Fig. 17c), and Model 4 employed 16 variables to predict elongation (Fig. 17d). Additionally, Model 5 utilized 26 variables to estimate impact strength (Fig. 17e).

These fuzzy logic models collectively generated 27 distinct rules. The Mamdani fuzzy inference system utilized these rules to predict mechanical properties such as tensile strength, hardness, flexural strength, elongation, impact strength, and modulus. The predictions were obtained through the defuzzification process using the centroid method. The development of the fuzzy logic models involved training them on 80% of the experimental data, while the remaining 20% was reserved for testing and validation. Detailed results of the fuzzy modeling process can be found in Sect. 3.2.

### 3.3 Multi objective optimization

Multi-objective optimization involves finding a set of solutions that simultaneously optimize multiple conflicting objectives. In this subsection, we explore the application of a desirability function in conjunction with the NSGA-II algorithm to tackle multi-objective optimization challenges. By combining the strengths of the desirability function, which allows for the incorporation of domain-specific preferences, and the NSGA-II algorithm, renowned for its ability to efficiently search for Pareto-optimal solutions, we aim to achieve superior results in optimizing multiple objectives. This subsection delves into the methodology and workflow of this integrated approach, highlighting its benefits and potential applications in diverse fields.

#### 3.3.1 Desirability function development

The desirability function is a commonly utilized tool for optimizing multiple objectives simultaneously. It is structured in a manner that when the objective is aimed at maximizing, the following relationship holds true:

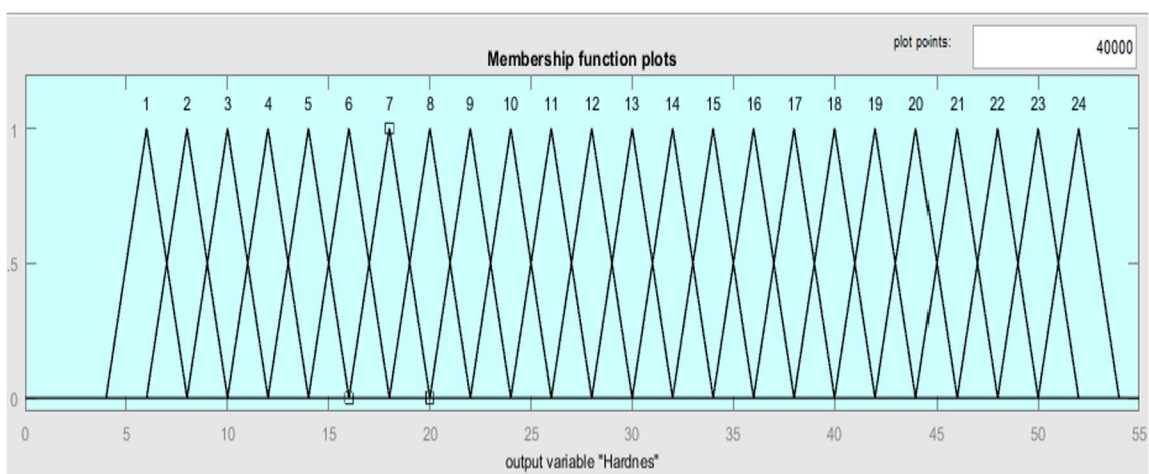
$$di = \begin{cases} 0 & \hat{y} \leq Li \\ \left( \frac{\hat{y} - Li}{Ti - Li} \right)^s & Li < \hat{y} < Ti \\ 1 & \hat{y} \geq Ti \end{cases} \quad (7)$$

Similarly, when the objective is to minimize, the following relationship holds true:

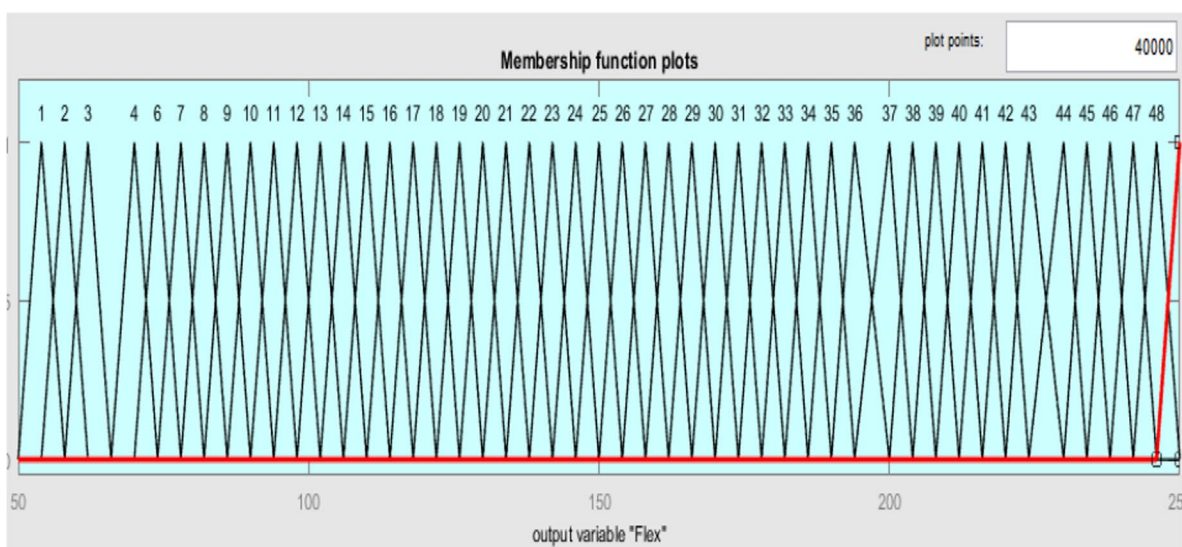
$$di = \begin{cases} 1 & \hat{y} \leq Ti \\ \left( \frac{Ui - \hat{y}}{Ui - Ti} \right)^s & Ti < \hat{y} < Ui \\ 0 & \hat{y} \geq Ui \end{cases} \quad (8)$$

If the desired response is a target value with upper and lower limits, the desirability can be expressed as follows:

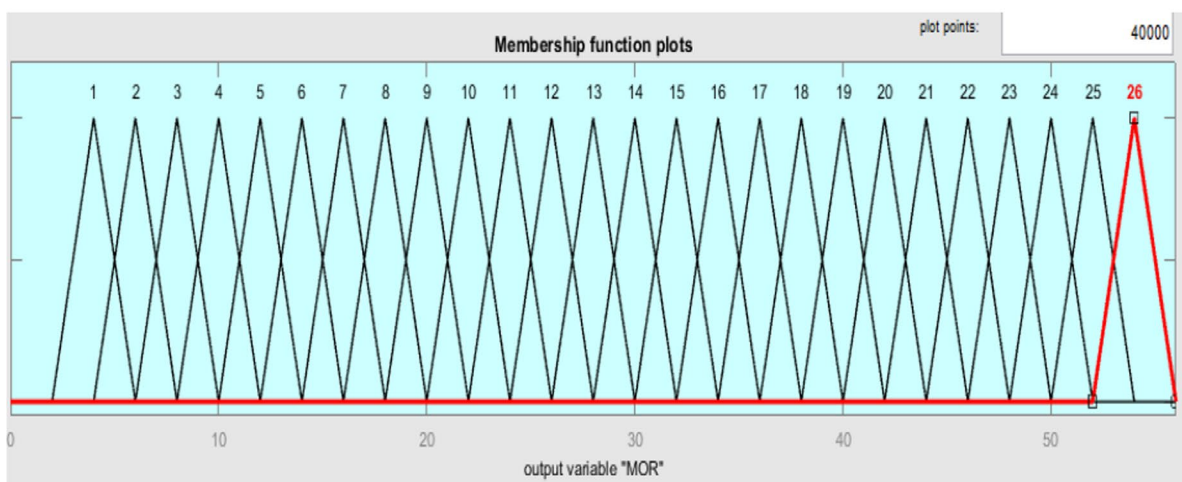
$$di = \left( \frac{\hat{y} - Li}{Ti - Li} \right)^s \quad Li \leq \hat{y} \leq Ti \quad (9)$$



**a: Membership function for hardness**

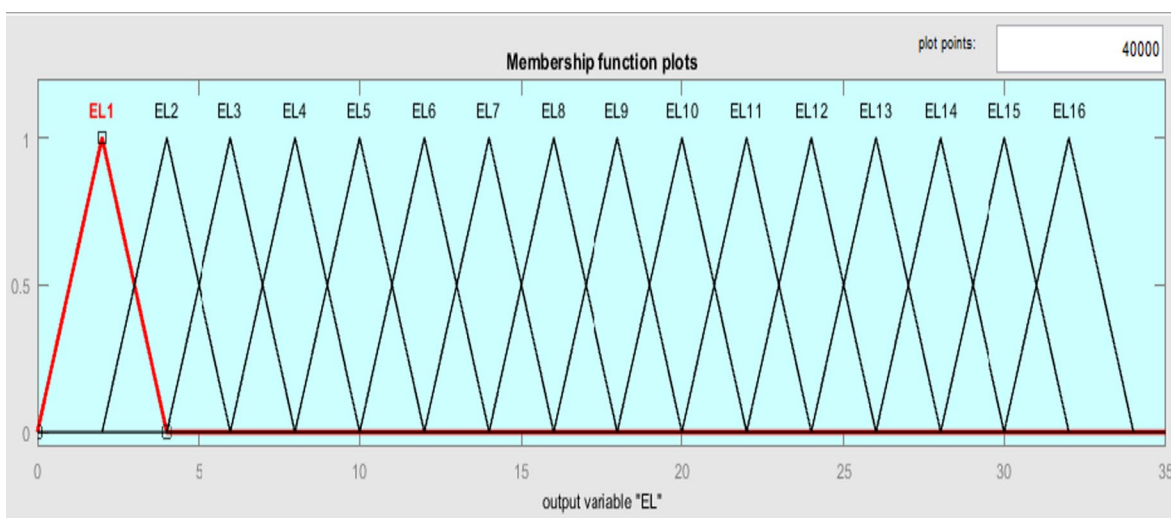


**b: Membership function for flexural strength**

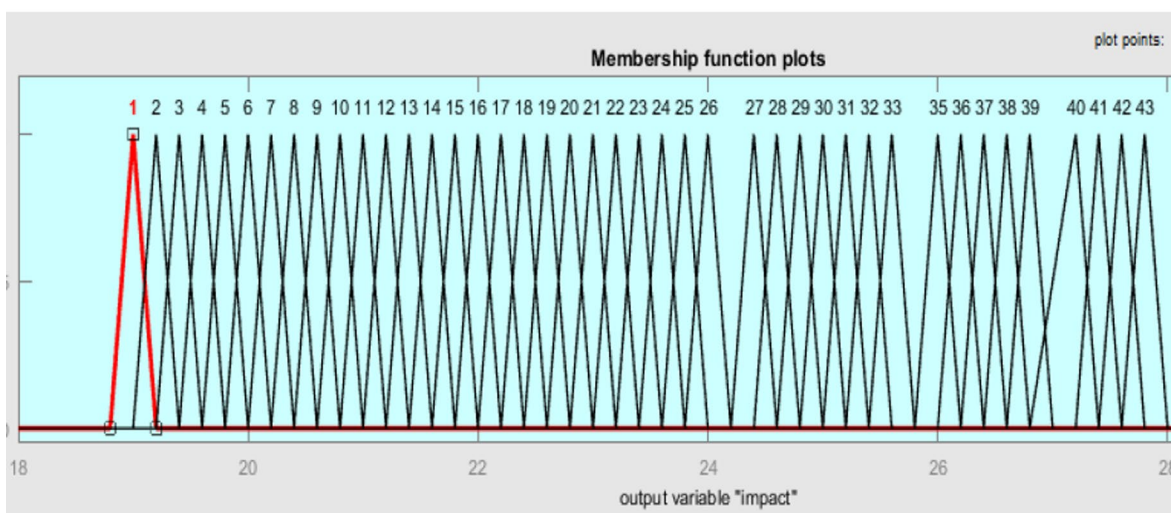


**c: Membership function for Modulus**

**Fig. 17** **a** Membership function for hardness. **b** Membership function for flexural strength. **c** Membership function for Modulus. **d** Membership function for Elongation. **e** Membership functions for impact strength



**d: Membership function for Elongation**



**e: Membership functions for impact strength.**

**Fig. 17** (continued)

$$di = \left( \frac{Ui - \hat{y}}{Ui - Ti} \right)^s \quad Ti \leq \hat{y} \leq Ui \quad (10)$$

$$di = 0 \quad \text{if } \hat{y} < Li \quad (11)$$

$$di = 0 \quad \text{if } \hat{y} > Ui \quad (12)$$

We introduce a new criterion where, if the desirability remains constant within a specific interval, the following equation is applicable:

$$di = \begin{cases} 0 & \hat{y} \leq Li \\ 1 & Li < \hat{y} < Ui \\ 0 & \hat{y} \geq Ui \end{cases} \quad (13)$$

**Table 6** AHP matrix

| Criteria           | Tensile Strength | Specific Strength | Flexural Strength | Elongation | Hardness | Impact Strength | Modulus | Specific Stiffness |
|--------------------|------------------|-------------------|-------------------|------------|----------|-----------------|---------|--------------------|
| Tensile Strength   | 1                | 1                 | 2                 | 3          | 4        | 5               | 6       | 6                  |
| Specific Strength  | 1                | 1                 | 2                 | 3          | 4        | 5               | 6       | 6                  |
| Flexural Strength  | 1/2              | 1/2               | 1                 | 2          | 3        | 4               | 5       | 5                  |
| Elongation         | 1/3              | 1/3               | 1/2               | 1          | 2        | 3               | 4       | 4                  |
| Hardness           | 1/4              | 1/4               | 1/3               | 1/2        | 1        | 2               | 3       | 3                  |
| Impact Strength    | 1/5              | 1/5               | 1/4               | 1/3        | 1/2      | 1               | 2       | 2                  |
| Modulus            | 1/6              | 1/6               | 1/5               | 1/4        | 1/3      | 1/2             | 1       | 1                  |
| Specific Stiffness | 1/6              | 1/6               | 1/5               | 1/4        | 1/3      | 1/2             | 1       | 1                  |

$$d_i = \begin{cases} \left( \frac{\hat{y} - L_i}{T_{1i} - L_i} \right)^s & \hat{y} \leq L_i \\ 1 & T_{1i} < \hat{y} < T_{2i} \\ \left( \frac{U_i - \hat{y}}{U_i - T_{2i}} \right)^s & \hat{y} \geq U_i \end{cases} \quad (14)$$

where  $d_i$  = desirability value;  $L_i$  is the lower limit of the response;  $U_i$  is the upper limit of the response;  $T_i$  is the target value;  $\hat{y}$  is the lower limit of the response.

The variable "s" represents a constant that determines the shape of the desirability function. When  $s = 1$ , the desirability function is linear. If  $s < 1$ , the desirability function is convex, and for  $s > 1$ , the desirability function is concave.

The composite desirability (DC) is calculated as follows:

$$DC = (d_1 \times d_2 \times d_3 \times \dots \times d_n)^{\frac{1}{n}} = \left( \prod_{i=1}^n d_i \right)^{\frac{1}{n}} \quad (15)$$

where  $n$  is the number of variables.

When weights are assigned to each  $d_i$ , denoted as  $w_i$ , the weights are defined such that:

$$w_1 + w_2 + w_3 \dots w_n = 1 \quad (16)$$

where  $0 < w_i < 1$ .

The determination of  $w_i$  can be achieved through various methods such as principal component analysis, rank order centroid method, analytic hierarchical process (AHP), entropy, and more.

$$DC = (d_1^{w_1} \times d_2^{w_2} \times d_3^{w_3} \times \dots \times d_n^{w_n})^{\frac{1}{n}} \quad (17)$$

The weights  $w_i$  were derived using the Analytic Hierarchy Process (AHP) methodology. The AHP matrix, derived from the attributes, is presented in Table 6.

Table 7 presents the priority vector, derived from the normalized eigenvector of the matrix shown in Table 6.

The parameters were assigned weights based on the priority vector as follows: 0.26135 for tensile strength, 0.26135 for specific strength, 0.166618 for flexural strength, 0.108597 for elongation, 0.071001 for hardness, 0.046807 for impact strength, and 0.032083 for specific stiffness. The optimization program's parameter settings were established according to the details provided in Table 8.

### 3.3.2 The codes for the new model

The algorithm below outlines the pseudocode for the implementation of the new desirability function criteria in the multi-objective optimization program developed for this research:

**Table 7** Priority vector matrix

| Criteria           | Tensile Strength | Specific Strength | Flexural Strength | Elongation | Hardness | Impact Strength | Modulus | Specific Stiffness | Priority |
|--------------------|------------------|-------------------|-------------------|------------|----------|-----------------|---------|--------------------|----------|
| Tensile Strength   | 1                | 1                 | 2                 | 3          | 4        | 5               | 6       | 6                  | 0.260276 |
| Specific Strength  | 1                | 1                 | 2                 | 3          | 4        | 5               | 6       | 6                  | 0.260276 |
| Flexural Strength  | 1/2              | 1/2               | 1                 | 2          | 3        | 4               | 5       | 5                  | 0.171214 |
| Elongation         | 1/3              | 1/3               | 1/2               | 1          | 2        | 3               | 4       | 4                  | 0.114833 |
| Hardness           | 1/4              | 1/4               | 1/3               | 1/2        | 1        | 2               | 3       | 3                  | 0.076688 |
| Impact Strength    | 1/5              | 1/5               | 1/4               | 1/3        | 1/2      | 1               | 2       | 2                  | 0.050608 |
| Modulus            | 1/6              | 1/6               | 1/5               | 1/4        | 1/3      | 1/2             | 1       | 1                  | 0.033053 |
| Specific Stiffness | 1/6              | 1/6               | 1/5               | 1/4        | 1/3      | 1/2             | 1       | 1                  | 0.033053 |

```
// First Pseudo code (Used in the desirability function algorithm)
```

```
if y >= Li && y <= Ui
```

```
    d(i) = 1
```

```
else
```

```
    d(i) = 0
```

```
end
```

```
// Second Pseudo code (Used in the desirability function algorithm)
```

```
if y >= T1 && y <= T2
```

```
    d(i) = 1
```

```
elseif y > T2
```

```
    d(i) = (Ui - y) / (Ui - T2)
```

```
elseif y > Ui
```

```
    d(i) = 0
```

```
elseif y < T2
```

```
    d(i) = (y - Li) / (T1 - Li)
```

```
else
```

```
    d(i) = 0
```

```
end
```

```
// Third Pseudo code (Used in NSGA-II)
```

```
if y >= T1 && y <= T2
```

```
    output = constant
```

```
elseif y > T2
```

```
    output = Ui - y
```

```
else
```

```
    output = y
```

```
end
```

### 3.3.3 Optimization of desirability function using fuzzy inference and PSO

The integration of a Fuzzy Inference System with a PSO optimized desirability function is illustrated in the program flowchart depicted in Fig. 18. This flowchart outlines the sequential steps involved in this integration process. Initially, the PSO generates an input denoted as X, which is a 1 by 3 matrix comprising three inputs: the weight percentage of bagasse, the weight percentage of sponge gourd, and the fiber size in micrometers. The PSO module then establishes communication with the DF module and transfers the input X to it.

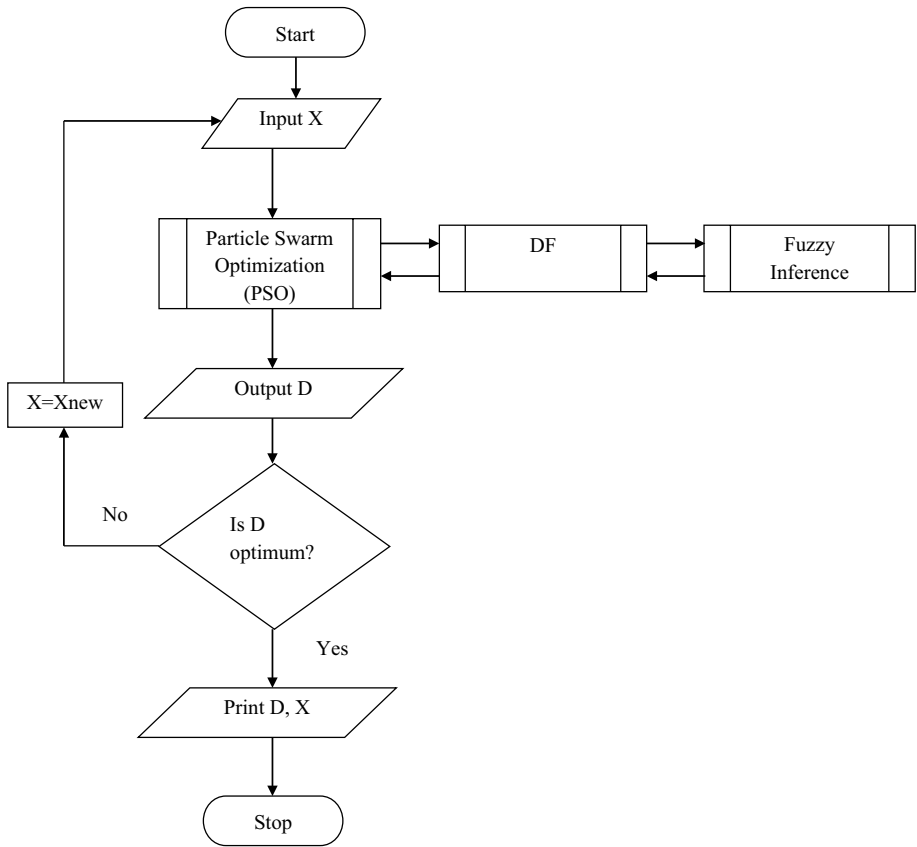
**Table 8** Parameter settings

| Parameter           | Lower limit | Upper limit | Weight   |
|---------------------|-------------|-------------|----------|
| <i>Inputs</i>       |             |             |          |
| %Wt of baggase      | 5           | 20          |          |
| %Wt of sponge gourd | 5           | 20          |          |
| Fiber size (μm)     | 150         | 600         |          |
| <i>Responses</i>    |             |             |          |
| tensile strength    | 50          | 250         | 0.260276 |
| Specific strength   | 40          | 300         | 0.260276 |
| Flexural strength   | 150         | 400         | 0.171214 |
| Elongation          | 10          | 20          | 0.114833 |
| Hardness            | 15          | 40          | 0.076688 |
| Impact strength     | 15          | 40          | 0.050608 |
| Modulus             | 30          | 100         | 0.033053 |
| Specific Stiffness  | 73          | 400         | 0.033053 |

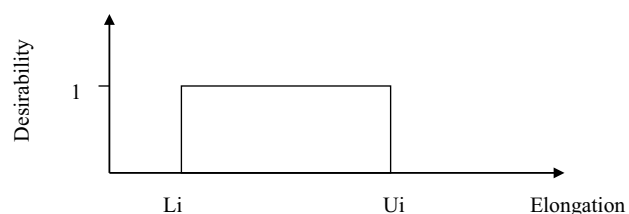
After receiving the input  $X$ , the DF module engages with the FIS module to derive the corresponding outputs, which include tensile strength, hardness, flexural strength, modulus, elongation, and impact strength. The DF module employs these outputs to calculate the desirability ( $D$ ) of the overall output. This iterative process persists until the optimum desirability ( $D$ ) is achieved.

To accommodate the distinct elongation property of the golf club material, specific adjustments were implemented in the desirability function model. These modifications were incorporated to ensure that the desirability aligns accurately with the graphical representation depicted in Fig. 19.

**Fig. 18** Flowchart of Fuzzy Inference coupled with PSO optimized desirability function



**Fig. 19** Desirability function graph for model 1



### 3.3.4 Fuzzy inference coupled NSGA-II model

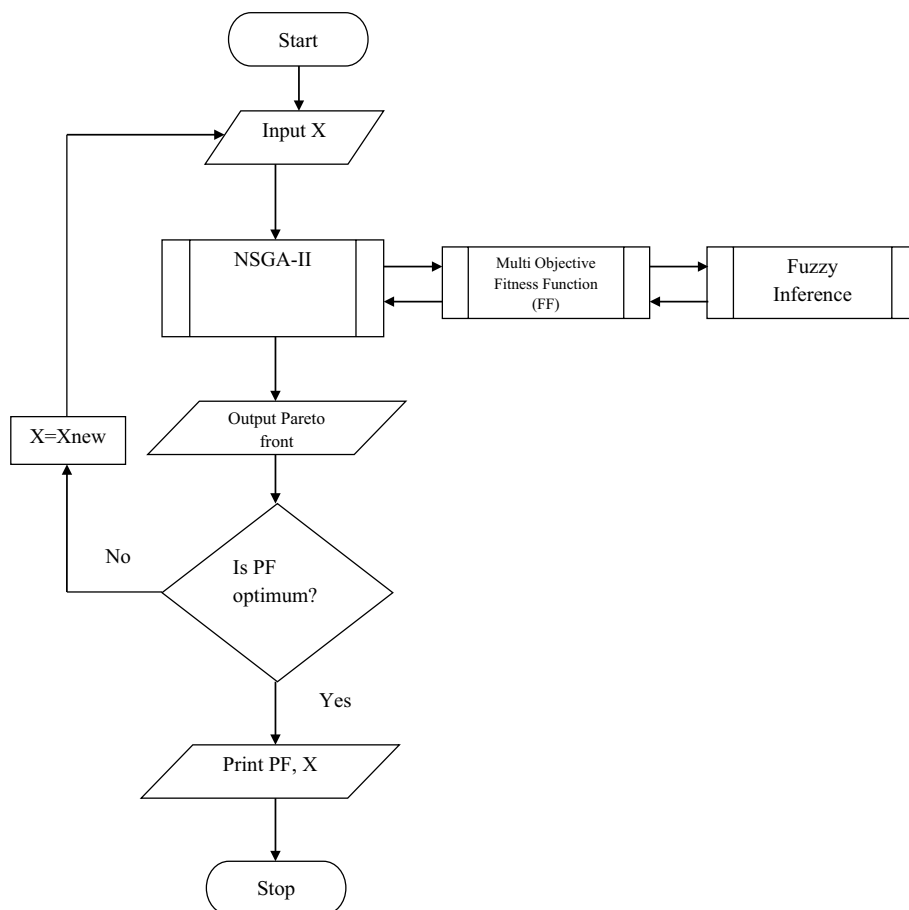
Figure 20 depicts a program flowchart showcasing the integration of a Fuzzy Inference System with the NSGA-II module. The flowchart provides a visual representation of the step-by-step process involved in this integration.

The NSGA-II module initiates the generation of an input labeled as X, which represents the initial population of input variables comprising the wt.% of bagasse, the wt.% of sponge gourd, and the fiber size in micrometers. Following this, the NSGA-II module establishes communication with the Multi-Objective Fitness Function (FF) module and transfers the input X to it. Upon receiving X, the FF module interacts with the Fuzzy Inference System (FIS) module to acquire the corresponding outputs, namely tensile strength, hardness, flexural strength, modulus, elongation, and impact strength, as mentioned earlier. The FF module utilizes the outputs obtained from the FIS module to evaluate the fitness of the input.

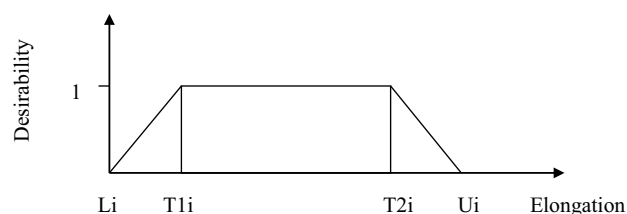
The iterative process persists until the optimum Pareto front is attained, which encompasses the set of optimal solutions where enhancing one objective necessitates compromising another. The flowchart visually depicts the sequence of steps executed by the Fuzzy Inference System in conjunction with the NSGA-II module to accomplish the desired outcome.

The elongation property, within the framework of the NSGA-II algorithm, is depicted and modeled through the utilization of Fig. 21. However, in contrast to the conventional implementation of desirability in the NSGA-II algorithm, the

**Fig. 20** Flowchart illustrating the program flow of the Fuzzy Inference coupled with NSGA-II



**Fig. 21** Desirability function graph for model 2



actual elongation values are directly employed to simulate desirability. In this simulation, higher values falling below the acceptable range are considered more favorable. Conversely, values within the acceptable range are deemed equally desirable. On the other hand, lower values exceeding the acceptable range are considered more preferable.

### 3.4 Summary of composite development and optimization process

Figure 22 illustrates the development and optimization process of the composite material. As shown in Fig. 22a, plant materials such as sponge gourd and bagasse were collected, cut, and processed into fibers. These fibers were then blended with epoxy resin, serving as the matrix, to form composite materials with varying compositions of fiber size (FS), weight percent of bagasse, and weight percent of sponge gourd. An experiment was designed using the Taguchi robust design method to determine variations in tensile strength (TS), hardness (H), flexural strength (Fst), modulus of elasticity (MOD), elongation (EI), impact strength (IS), specific strength (SStr), and specific stiffness (Sstif).

The experimental data collected was used to train a fuzzy logic (FL) model. This FL model was integrated with a desirability function program, which was programmed using weights derived from the Analytic Hierarchy Process (AHP). The output from the desirability function program was fed into a Particle Swarm Optimization (PSO) algorithm to obtain the optimum values for the desirability and the associated input and response variables of the composite material.

Additionally, the FL model was used as the fitness function in the NSGA-II optimization algorithm, as depicted in Fig. 22b, to achieve optimal values on the Pareto front.

## 4 Results and discussion

### 4.1 Experimental results

Table 9 presents the experimental results obtained for determining the mechanical properties of the composite material, as outlined in Sect. 3.3. A series of 27 experimental runs were performed, leading to the following observations:

- Tensile Strength: The maximum value recorded was 80 MPa, while the minimum value obtained was 26.5 MPa.
- Hardness: The maximum recorded value was 52, with a corresponding minimum value of 10.
- Flexural Strength: The maximum value observed was 240 MPa, while the minimum value was 55 MPa.
- Modulus: The maximum value recorded for the modulus was 50 GPa, and the minimum value was 9 GPa.
- Elongation: The maximum value obtained for elongation was 33, while the minimum value measured was 11.
- Impact Strength: The maximum value recorded for impact strength was 27.44 MPa, and the minimum value was 19 MPa.

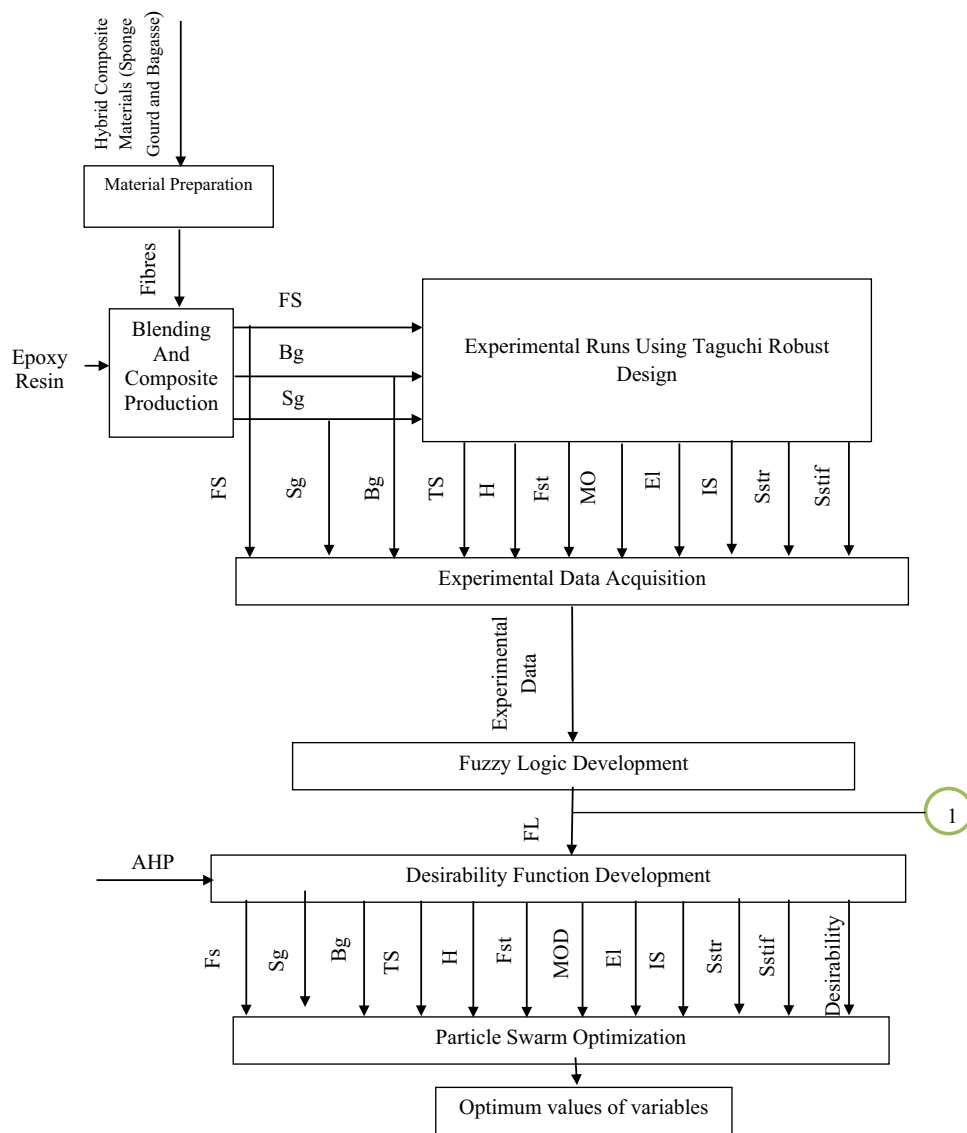
These findings provide quantitative data on the mechanical properties of the composite material, offering insights into the range of values observed for each property during the experiment.

The recorded property values fall within the specified range outlined in Table 10, which represents the required properties for golf club materials.

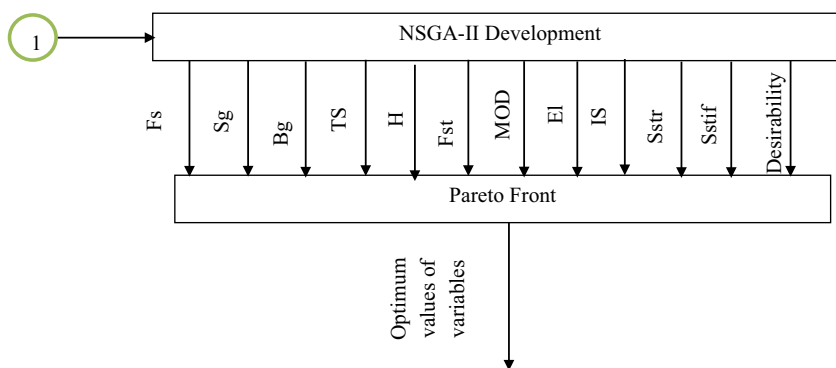
### 4.2 Fuzzy logic modelling results

Table 11 displays the recorded experimental values alongside the fuzzy logic predicted values of the mechanical properties for the hybrid polymer composite material.

**Fig. 22** Composite development and optimization process. **a** Composite development and optimization procedure I. **b** Composite development and optimization procedure II



(a) Composite development and optimization procedure I



(b) Composite development and optimization procedure II

**Table 9** Experimental results

| S/N | %Wt of baggase | %Wt of<br>Sponge<br>gourd | Fiber size (μm) | Tensile<br>strength<br>(MPa) | Hardness<br>(Rockwell) | Flexural<br>strength<br>(MPa) | Modulus (GPa) | Elongation | Impact<br>strength<br>(MPa) |
|-----|----------------|---------------------------|-----------------|------------------------------|------------------------|-------------------------------|---------------|------------|-----------------------------|
| 1   | 5.00           | 5.00                      | 600.00          | 32.00                        | 17.90                  | 90.00                         | 10.00         | 12.00      | 19.00                       |
| 2   | 10.00          | 5.00                      | 600.00          | 35.00                        | 18.20                  | 103.00                        | 13.50         | 16.00      | 20.58                       |
| 3   | 20.00          | 5.00                      | 600.00          | 42.00                        | 18.40                  | 110.00                        | 15.00         | 18.00      | 22.00                       |
| 4   | 5.00           | 10.00                     | 600.00          | 33.00                        | 18.60                  | 90.00                         | 22.00         | 20.00      | 22.54                       |
| 5   | 10.00          | 10.00                     | 600.00          | 33.00                        | 20.00                  | 85.00                         | 25.00         | 22.00      | 21.56                       |
| 6   | 20.00          | 10.00                     | 600.00          | 31.00                        | 17.90                  | 80.00                         | 12.00         | 26.00      | 22.00                       |
| 7   | 5.00           | 20.00                     | 600.00          | 33.60                        | 41.00                  | 98.00                         | 18.00         | 33.00      | 19.60                       |
| 8   | 10.00          | 20.00                     | 600.00          | 30.00                        | 37.00                  | 65.00                         | 40.00         | 22.00      | 21.56                       |
| 9   | 20.00          | 20.00                     | 600.00          | 27.00                        | 49.00                  | 55.00                         | 45.00         | 21.00      | 19.00                       |
| 10  | 5.00           | 5.00                      | 300.00          | 38.00                        | 31.00                  | 105.00                        | 22.70         | 17.00      | 19.60                       |
| 11  | 10.00          | 5.00                      | 300.00          | 45.00                        | 28.00                  | 115.00                        | 24.00         | 16.00      | 20.00                       |
| 12  | 20.00          | 5.00                      | 300.00          | 54.00                        | 26.00                  | 120.00                        | 25.00         | 13.00      | 21.56                       |
| 13  | 5.00           | 10.00                     | 300.00          | 39.00                        | 25.50                  | 95.00                         | 14.00         | 13.00      | 21.56                       |
| 14  | 10.00          | 10.00                     | 300.00          | 60.00                        | 30.00                  | 156.00                        | 25.00         | 12.00      | 27.44                       |
| 15  | 20.00          | 10.00                     | 300.00          | 41.00                        | 33.00                  | 110.00                        | 22.00         | 12.00      | 21.00                       |
| 16  | 5.00           | 20.00                     | 300.00          | 38.00                        | 40.00                  | 100.00                        | 20.00         | 11.00      | 19.60                       |
| 17  | 10.00          | 20.00                     | 300.00          | 27.00                        | 38.00                  | 80.00                         | 13.00         | 13.00      | 26.46                       |
| 18  | 20.00          | 20.00                     | 300.00          | 26.50                        | 36.00                  | 78.00                         | 17.00         | 18.00      | 20.58                       |
| 19  | 5.00           | 5.00                      | 150.00          | 49.00                        | 48.50                  | 135.00                        | 26.50         | 12.00      | 20.58                       |
| 20  | 10.00          | 5.00                      | 150.00          | 75.00                        | 51.00                  | 215.00                        | 9.00          | 14.00      | 20.58                       |
| 21  | 20.00          | 5.00                      | 150.00          | 63.00                        | 51.50                  | 170.00                        | 47.00         | 12.00      | 20.00                       |
| 22  | 5.00           | 10.00                     | 150.00          | 45.00                        | 52.00                  | 119.00                        | 16.00         | 23.00      | 20.58                       |
| 23  | 10.00          | 10.00                     | 150.00          | 80.00                        | 20.00                  | 240.00                        | 50.00         | 12.00      | 20.58                       |
| 24  | 20.00          | 10.00                     | 150.00          | 75.00                        | 18.00                  | 210.00                        | 40.00         | 12.00      | 20.20                       |
| 25  | 5.00           | 20.00                     | 150.00          | 68.00                        | 18.00                  | 202.00                        | 41.50         | 12.00      | 20.58                       |
| 26  | 10.00          | 20.00                     | 150.00          | 62.00                        | 13.00                  | 171.00                        | 47.40         | 12.00      | 23.52                       |
| 27  | 20.00          | 20.00                     | 150.00          | 45.50                        | 10.00                  | 140.00                        | 20.40         | 20.00      | 21.56                       |

**Table 10** The properties of golf club materials

| Properties                    | Value   |
|-------------------------------|---------|
| 1. Tensile strength (MPa)     | 50–250  |
| 2. Flexural strength (MPa)    | 150–400 |
| 3. Hardness (Rockwell)        | 15–40   |
| 4. Elongation (%)             | 10–20   |
| 5. Specific strength (Nm/kg)  | 15–40   |
| 6. Specific stiffness (Nm/kg) | 20–50   |
| 7. Impact strength (MPa)      | 15–40   |
| 8. Modulus (GPa)              | 30–100  |

Table 12 displays the correlation coefficients between the experimentally determined values and the fuzzy logic predicted values presented in Table 11. Analyzing the data in Table 12 reveals that the flexural strength exhibits the highest correlation coefficient (R) value of 0.9922, while the impact strength has the lowest correlation coefficient value of 0.9529. Furthermore, Table 12 includes a comparison between the fuzzy logic predictions and the predictions made by an Artificial Neural Network (ANN) model. Overall, the fuzzy logic predictions demonstrate slightly superior performance compared to the ANN predictions. As a result, the fuzzy logic approach was chosen for the development of the models in this study.

Table 11 Experimental results and fuzzy logic predictions

| Tensile strength | Predicted ten-<br>sile strength | Hardness | Predicted<br>hardness | Flexural strength | Predicted flex-<br>ural strength | Modulus | Predicted<br>modulus | Elongation | Predicted<br>Elongation | Impact Strength | Predicted<br>impact<br>strength |
|------------------|---------------------------------|----------|-----------------------|-------------------|----------------------------------|---------|----------------------|------------|-------------------------|-----------------|---------------------------------|
| 95.00            | 94.00                           | 17.90    | 18.00                 | 90.00             | 90.00                            | 10.00   | 10.00                | 12.00      | 12.00                   | 19.00           | 19.00                           |
| 103.60           | 102.00                          | 18.20    | 18.00                 | 103.00            | 102.00                           | 13.50   | 14.00                | 16.00      | 16.00                   | 20.58           | 20.60                           |
| 120.00           | 118.00                          | 18.40    | 18.00                 | 110.00            | 110.00                           | 15.00   | 14.00                | 18.00      | 18.00                   | 22.00           | 22.00                           |
| 100.00           | 98.00                           | 18.60    | 18.00                 | 90.00             | 90.00                            | 22.00   | 22.00                | 20.00      | 20.00                   | 22.54           | 22.60                           |
| 100.00           | 98.00                           | 20.00    | 20.00                 | 85.00             | 86.00                            | 25.00   | 24.00                | 22.00      | 22.00                   | 21.56           | 21.60                           |
| 95.00            | 94.00                           | 17.90    | 18.00                 | 80.00             | 78.00                            | 12.00   | 12.00                | 26.00      | 26.00                   | 22.00           | 22.00                           |
| 102.00           | 102.00                          | 41.00    | 40.00                 | 98.00             | 98.00                            | 18.00   | 18.00                | 33.00      | 32.00                   | 19.60           | 19.60                           |
| 70.00            | 70.00                           | 37.00    | 36.00                 | 65.00             | 70.00                            | 40.00   | 40.00                | 22.00      | 22.00                   | 21.56           | 21.60                           |
| 60.00            | 58.00                           | 49.00    | 49.00                 | 55.00             | 54.00                            | 45.00   | 44.00                | 21.00      | 20.00                   | 19.00           | 19.00                           |
| 107.00           | 106.00                          | 31.00    | 30.00                 | 105.00            | 106.00                           | 22.70   | 22.00                | 17.00      | 16.00                   | 19.60           | 19.60                           |
| 130.00           | 130.00                          | 28.00    | 28.00                 | 115.00            | 114.00                           | 24.00   | 24.00                | 16.00      | 16.00                   | 20.00           | 20.00                           |
| 150.00           | 150.00                          | 26.00    | 26.00                 | 120.00            | 118.00                           | 25.00   | 24.00                | 13.00      | 12.00                   | 21.56           | 21.60                           |
| 110.00           | 110.00                          | 25.50    | 26.00                 | 95.00             | 94.00                            | 14.00   | 14.00                | 13.00      | 12.00                   | 21.56           | 21.60                           |
| 170.00           | 170.00                          | 30.00    | 30.00                 | 156.00            | 154.00                           | 25.00   | 32.00                | 12.00      | 12.00                   | 27.44           | 27.40                           |
| 115.00           | 114.00                          | 33.00    | 32.00                 | 110.00            | 110.00                           | 22.00   | 22.00                | 12.00      | 12.00                   | 21.00           | 21.00                           |
| 110.00           | 110.00                          | 40.00    | 40.00                 | 100.00            | 98.00                            | 20.00   | 20.00                | 11.00      | 10.00                   | 19.60           | 19.60                           |
| 83.00            | 82.00                           | 38.00    | 38.00                 | 80.00             | 78.00                            | 13.00   | 12.00                | 13.00      | 12.00                   | 26.46           | 26.40                           |
| 81.00            | 82.00                           | 36.00    | 36.00                 | 78.00             | 78.00                            | 17.00   | 16.00                | 18.00      | 18.00                   | 20.58           | 20.60                           |
| 138.00           | 148.00                          | 48.50    | 48.00                 | 135.00            | 136.00                           | 26.50   | 27.00                | 12.00      | 12.00                   | 20.58           | 20.60                           |
| 216.60           | 216.00                          | 51.00    | 50.00                 | 215.00            | 216.00                           | 49.00   | 50.00                | 14.00      | 14.00                   | 20.58           | 20.60                           |
| 180.00           | 182.00                          | 51.50    | 52.00                 | 170.00            | 170.00                           | 47.00   | 46.00                | 12.00      | 12.00                   | 20.00           | 20.00                           |
| 121.00           | 122.00                          | 52.00    | 52.00                 | 119.00            | 118.00                           | 16.00   | 16.00                | 23.00      | 22.00                   | 20.58           | 20.60                           |
| 248.00           | 246.00                          | 20.00    | 20.00                 | 240.00            | 238.00                           | 50.00   | 50.00                | 12.00      | 12.00                   | 20.58           | 20.60                           |
| 220.00           | 220.00                          | 18.00    | 18.00                 | 210.00            | 212.00                           | 40.00   | 40.00                | 12.00      | 12.00                   | 20.20           | 20.20                           |
| 206.00           | 204.00                          | 18.00    | 18.00                 | 202.00            | 199.00                           | 41.50   | 42.00                | 12.00      | 12.00                   | 20.58           | 20.60                           |
| 181.00           | 182.00                          | 13.00    | 12.00                 | 171.00            | 170.00                           | 47.40   | 48.00                | 12.00      | 12.00                   | 23.52           | 23.40                           |
| 144.00           | 142.00                          | 10.00    | 9.90                  | 140.00            | 138.00                           | 20.40   | 20.00                | 20.00      | 20.00                   | 21.56           | 21.60                           |

**Table 12** Statistics of fuzzy logic predictions of responses

| Mechanical property | Fuzzy logic predictions vs experimental data (R value) | ANN predictions vs Experimental Data (R value) |
|---------------------|--------------------------------------------------------|------------------------------------------------|
| Tensile Strength    | 0.9912                                                 | 0.9507                                         |
| Hardness            | 0.9657                                                 | 0.9699                                         |
| Flexural Strength   | 0.9922                                                 | 0.9544                                         |
| Modulus             | 0.9857                                                 | 0.9685                                         |
| Elongation          | 0.9637                                                 | 0.9642                                         |
| Impact Strength     | 0.9529                                                 | 0.9684                                         |

The implications of the provided results are significant for several reasons. Firstly, the correlation coefficients presented in Table 12 provide insights into the relationship between the experimentally determined values and the fuzzy logic predicted values from Table 11. A high correlation coefficient indicates a strong linear relationship between the two sets of values, suggesting that the fuzzy logic model effectively captures the patterns and trends present in the experimental data. In this case, the flexural strength exhibits the highest correlation coefficient value of 0.9912, indicating a very strong agreement between the predicted and experimental values for this particular mechanical property. On the other hand, the impact strength shows a slightly lower correlation coefficient value of 0.9529, suggesting a slightly weaker but still significant relationship between the predicted and experimental values for this property.

Furthermore, the comparison between the fuzzy logic predictions and predictions made by an Artificial Neural Network (ANN) model, as presented in Table 12, is noteworthy. The slightly superior performance of the fuzzy logic predictions suggests that the fuzzy logic model, with its ability to handle imprecise and uncertain data, is well-suited for accurately forecasting and optimizing the mechanical properties of the hybrid composite material under investigation. This finding implies that fuzzy logic can offer advantages over ANN models in this particular context, potentially leading to more reliable and accurate predictions.

Overall, these implications highlight the efficacy of the fuzzy logic model, in combination with the correlation coefficients and the comparison with the ANN model, in accurately predicting and optimizing the mechanical properties of the hybrid composite material. These findings contribute to the advancement of material forecasting and optimization processes, enabling researchers and engineers to make informed decisions and design materials with improved mechanical performance.

### 4.3 Optimization results

Following the execution of the modified NSGA-II algorithm, Table 13 presents the process decision variables associated with each Pareto optimal solution. Upon careful examination of Table 13, it becomes evident that only 5 parameters on the Pareto front exhibit desirability values above zero. This indicates that the remaining process variables on the front fail to meet the required mechanical properties for golf club applications, as outlined in Table 10.

Table 14 showcases the optimal solution attained after running the modified desirability function program, as depicted in Fig. 18's flowchart. Through the optimization algorithm, an optimum desirability value of 0.8418 was obtained across three runs. The distinctive aspect of the desirability function model is the introduced modification in Eq. 15, which accurately represents the elongation property.

When designing such properties, designers typically adopt either the "larger the better" or the "nominal the best" model assumption. If the "larger the better" model (Eq. 9) was employed for elongation, the results presented in Table 15 were observed. Similarly, adopting the "nominal the best" model (Eqs. 11–14) yielded the outcomes shown in Table 16. Comparing these results with those obtained from the modified model, it is evident that the properties achieved through the modified model were superior. This is highlighted in Table 11, where the desirability values are notably higher than those in Tables 15 and 16.

The implications of the provided result are highly significant in terms of the suitability of the optimized composite material for golf club applications. By comparing the optimal mechanical properties presented in Table 14 with the specified mechanical properties outlined in Table 10, it becomes evident that the optimized composite material meets and even exceeds the designated requirements for golf club applications.

**Table 13** Process decision variables corresponding to each of the Pareto optimal solutions

| S/N | %Bg   | %SG   | FS     | TS    | H     | FSt    | Mod   | El    | IS    | Sstr  | Sstif | Des  |
|-----|-------|-------|--------|-------|-------|--------|-------|-------|-------|-------|-------|------|
| 1   | 5.00  | 5.00  | 150.00 | 53.63 | 39.37 | 160.50 | 26.58 | 15.98 | 20.61 | 33.52 | 16.61 | 0.00 |
| 2   | 13.80 | 10.00 | 168.03 | 81.01 | 19.98 | 237.98 | 50.01 | 11.97 | 20.63 | 50.63 | 31.26 | 0.83 |
| 3   | 12.59 | 10.00 | 168.03 | 81.00 | 19.98 | 237.99 | 49.99 | 11.99 | 20.61 | 50.63 | 31.25 | 0.83 |
| 4   | 10.00 | 15.98 | 202.93 | 27.40 | 37.94 | 78.00  | 11.87 | 11.95 | 26.36 | 17.13 | 7.42  | 0.00 |
| 5   | 13.67 | 10.00 | 168.04 | 81.02 | 19.99 | 237.99 | 50.00 | 11.98 | 20.63 | 50.64 | 31.25 | 0.83 |
| 6   | 10.00 | 10.17 | 281.57 | 58.09 | 30.51 | 150.19 | 30.72 | 11.99 | 27.33 | 36.31 | 19.20 | 0.00 |
| 7   | 9.75  | 16.58 | 504.17 | 32.34 | 38.16 | 85.99  | 25.75 | 25.50 | 20.61 | 20.21 | 16.09 | 0.00 |
| 8   | 15.75 | 17.41 | 512.55 | 27.19 | 48.00 | 54.00  | 44.01 | 19.98 | 18.97 | 16.99 | 27.51 | 0.00 |
| 9   | 12.96 | 16.31 | 502.64 | 28.90 | 41.70 | 58.00  | 42.00 | 21.00 | 20.28 | 18.06 | 26.25 | 0.00 |
| 10  | 10.00 | 15.87 | 156.49 | 63.00 | 12.01 | 170.00 | 48.01 | 11.99 | 23.40 | 39.38 | 30.00 | 0.00 |
| 11  | 15.43 | 16.70 | 503.82 | 27.37 | 47.85 | 54.00  | 44.12 | 19.95 | 18.96 | 17.10 | 27.57 | 0.00 |
| 12  | 14.83 | 16.17 | 506.36 | 27.85 | 45.14 | 55.67  | 43.08 | 20.36 | 19.51 | 17.40 | 26.93 | 0.00 |
| 13  | 10.00 | 12.84 | 247.91 | 42.33 | 34.39 | 112.41 | 21.14 | 11.99 | 26.83 | 26.46 | 13.21 | 0.00 |
| 14  | 5.13  | 10.48 | 153.63 | 51.99 | 37.04 | 143.46 | 24.68 | 21.51 | 20.82 | 32.50 | 15.42 | 0.00 |
| 15  | 9.94  | 11.99 | 172.37 | 71.88 | 17.17 | 205.80 | 47.53 | 12.66 | 21.91 | 44.93 | 29.71 | 0.79 |
| 16  | 14.85 | 15.86 | 270.90 | 27.07 | 36.07 | 78.00  | 15.78 | 17.67 | 20.91 | 16.92 | 9.86  | 0.00 |
| 17  | 15.43 | 16.71 | 503.81 | 27.37 | 47.85 | 54.00  | 44.12 | 19.95 | 18.96 | 17.10 | 27.57 | 0.00 |
| 18  | 9.91  | 9.94  | 177.24 | 75.51 | 22.66 | 224.61 | 45.77 | 13.03 | 20.61 | 47.20 | 28.61 | 0.82 |

**Table 14** Optimum solutions obtained using the modified desirability function

| Run | Bg% | SG%  | Fs     | TS    | H     | FSt    | Mod   | El    | Is    | Sstr  | Sstif | Desirability |
|-----|-----|------|--------|-------|-------|--------|-------|-------|-------|-------|-------|--------------|
| 1   | 10  | 6.62 | 150.00 | 78.16 | 39.33 | 227.97 | 49.99 | 12.94 | 20.61 | 48.85 | 24.58 | 0.84         |
| 2   | 10  | 6.62 | 150.00 | 78.16 | 39.33 | 227.97 | 49.99 | 12.94 | 20.61 | 48.85 | 24.58 | 0.84         |
| 3   | 10  | 6.62 | 150.00 | 78.16 | 39.33 | 227.97 | 49.99 | 12.94 | 20.61 | 48.85 | 24.58 | 0.84         |

**Table 15** Optimum solutions obtained using the larger the better assumption

| Run | Bg% | SG%  | Fs     | TS    | H     | FSt    | Mod   | El    | Is    | Sstr  | Sstif | Desirability |
|-----|-----|------|--------|-------|-------|--------|-------|-------|-------|-------|-------|--------------|
| 1   | 10  | 5.52 | 150.00 | 78.01 | 40.00 | 227.29 | 49.99 | 12.99 | 20.61 | 48.76 | 25.00 | 0.82         |
| 2   | 10  | 5.52 | 150.00 | 78.01 | 40.00 | 227.29 | 49.99 | 12.99 | 20.61 | 48.76 | 25.00 | 0.82         |
| 3   | 10  | 5.52 | 150.00 | 78.01 | 40.00 | 227.29 | 49.99 | 12.99 | 20.61 | 48.76 | 25.00 | 0.82         |

**Table 16** Optimum solutions obtained using the nominal the best assumption

| Run | Bg% | SG%   | Fs     | TS    | H     | FSt    | Mod   | El    | Is    | Sstr  | Sstif | Desirability |
|-----|-----|-------|--------|-------|-------|--------|-------|-------|-------|-------|-------|--------------|
| 1   | 10  | 5.523 | 150.00 | 78.01 | 39.99 | 227.30 | 49.99 | 12.99 | 20.61 | 48.76 | 25.00 | 0.83         |
| 2   | 10  | 5.523 | 150.00 | 78.01 | 39.99 | 227.30 | 49.99 | 12.99 | 20.61 | 48.76 | 25.00 | 0.83         |
| 3   | 10  | 5.523 | 150.00 | 78.01 | 39.99 | 227.30 | 49.99 | 12.99 | 20.61 | 48.76 | 25.00 | 0.83         |

This finding is crucial because it indicates that the composite material can be considered a highly promising candidate for manufacturing golf clubs. The fact that the optimized material closely aligns with the specified mechanical properties demonstrates that the material possesses the necessary characteristics to deliver the desired performance and functionality in golf clubs.

Furthermore, surpassing the designated requirements implies that the optimized composite material has the potential to offer enhanced performance and superior durability compared to conventional golf club materials. This has significant implications for golfers, as it suggests that the use of this composite material in club manufacturing could lead to improved gameplay, longer lifespan of the clubs, and potentially better overall results for golfers.

Moreover, meeting or surpassing the specified mechanical properties for golf club materials also signifies that the composite material has undergone a successful optimization process. This suggests that the methodology employed to optimize the material was effective in achieving the desired outcomes and can be applied in future material design and optimization projects.

Overall, the implications of this result are highly promising for the golf industry. The optimized composite material demonstrates excellent compatibility with the mechanical requirements of golf clubs, offering the potential for enhanced performance and durability. These findings provide valuable insights for manufacturers and designers, enabling them to select and utilize materials that can lead to the development of high-quality golf clubs that meet the specific demands of players.

## 5 Conclusion

The research successfully demonstrated the development and optimization of a hybrid composite material comprising sponge gourd, bagasse, and epoxy resin for potential use in golf club applications. The experimental phase, involving 27 runs, provided critical insights into the mechanical properties of the composite, yielding a range of tensile strength, hardness, flexural strength, modulus, elongation, and impact strength values. These experimental results, compared with the required properties for golf club materials, confirmed the composite's suitability within the specified ranges.

The FL model, incorporating three input factors and six response variables, accurately predicted the material's tensile strength, hardness, flexural strength, modulus, elongation, and impact strength. The FL modeling results indicated a high correlation between the predicted and experimentally determined mechanical properties, with the flexural strength showing the highest correlation coefficient (0.9922). The comparison between the FL model and the ANN model underscored the FL model's superior performance in handling imprecise data and accurately forecasting the composite's mechanical properties.

Optimization using the modified NSGA-II algorithm and a desirability function program demonstrated that only five parameters on the Pareto front exhibited desirability values above zero, aligning with the required mechanical properties for golf clubs. The optimization process using the modified desirability function yielded an optimum desirability value of 0.84, indicating the composite's potential to meet and exceed the specified requirements. In comparison with the normal "larger the better" or the "nominal the best" model assumption, the optimum desirability values were 0.82 and 0.83 respectively. The higher value of desirability obtained from the modified desirability function is indicative of the superiority of the model.

The integration of a modified DF algorithm and the NSGA-II algorithm further enhanced the optimization of these properties, with the modified algorithms effectively addressing specific characteristics such as elongation and producing superior desirability values. The results obtained emphasize the successful utilization of the FL model alongside the modified DF algorithm and NSGA-II in accurately forecasting and optimizing the newly formulated composite material's mechanical properties. These findings significantly contribute to the progression of material prediction and optimization procedures.

These findings suggest that the optimized composite material not only meets the necessary mechanical criteria for golf clubs but also offers enhanced performance and durability compared to conventional materials. The successful application of the FL model and the optimization methodology in this research highlights their effectiveness in material design and optimization projects, paving the way for future advancements in composite material development for various applications.

In conclusion, the optimized hybrid composite material presents a promising candidate for golf club manufacturing, with potential benefits including improved gameplay, extended club lifespan, and superior overall performance for golfers. This research provides valuable insights for manufacturers and designers, contributing to the development of high-quality, performance-enhancing golf clubs.

## 6 Future work

The study's conclusion highlights the successful utilization of fuzzy logic in predicting the mechanical properties of a newly developed hybrid composite material for golf club applications. Based on this conclusion, several possible future directions for research can be explored:

1. **Refinement and Expansion of the Fuzzy Logic Model:** Further refinement and enhancement of the fuzzy logic model can be pursued to improve its predictive capabilities. This may involve exploring different membership functions, fuzzy rules, or inference methods to achieve even higher accuracy in predicting the mechanical properties of composite materials. Furthermore, an exploration into the predictive capabilities of ANFIS model could provide valuable insights and strengthen its effectiveness as a prediction tool in this domain.
2. **Optimization Algorithm Exploration:** The study mentions the integration of a modified desirability function algorithm and the NSGA-II algorithm for optimizing the material's properties. Future research can focus on exploring other optimization algorithms and comparing their performance with the modified algorithms used in this study. This may involve investigating algorithms such as genetic algorithms, particle swarm optimization, or evolutionary algorithms to identify the most effective approach for optimizing composite material properties.
3. **Multi-Criteria Decision-Making (MCDM) Tools Exploration:** To advance future research in this area, we suggest exploring alternative multi-criteria decision-making (MCDM) tools like utility functions and TOPSIS, and optimizing them using diverse algorithms such as simulated annealing, ant colony optimization (ACO), or differential evolution.
4. **Validation and Experimental Verification:** While the study demonstrates successful predictions using the fuzzy logic model, further validation and experimental verification can be conducted to ensure the model's accuracy across different material formulations, processing conditions, and testing methods. Experimental studies can be designed to collect additional data to validate the predictions made by the fuzzy logic model.
5. **Material Characterization and Microstructural Analysis:** Future research can focus on the characterization and analysis of the hybrid composite material at a microstructural level. This may involve techniques such as scanning electron microscopy, X-ray diffraction, or spectroscopy to gain insights into the material's internal structure and composition, and how they relate to the predicted mechanical properties. Such investigations can provide a deeper understanding of the material's behavior and aid in further improving the predictive models.
6. **Performance Testing and Real-world Applications:** In addition to predicting mechanical properties, future research can involve testing the actual performance of the hybrid composite material in golf club applications. This may include conducting swing tests, durability assessments, or comparative studies with existing materials. Real-world applications can provide valuable insights into the material's suitability, performance, and potential for commercial use.

Overall, the study's conclusion sets the stage for future research to continue advancing the field of material prediction and optimization. By addressing the areas mentioned above, researchers can further refine predictive models, explore alternative optimization algorithms, validate predictions, analyze material microstructure, and evaluate real-world performance. These avenues of research will contribute to the continued development and practical implementation of hybrid composite materials in golf club applications, as well as in other industries that benefit from improved material prediction and optimization processes.

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## Declarations

**Competing interests** The authors declare no competing interests.

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